SeSPHR A Methodology for Secure Sharing of Personal Health Records in the Cloud

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of the requirements for the award of the degree

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In

COMPUTER SCIENCE AND ENGINEERING

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Maisammaguda, Medchal (Dist), Hyderabad -500100, Telangana.

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CERTIFICATE

This is to certify that the Project Stage-II entitled **"SeSPHR A Methodology for Secure Sharing of Personal Health Records in the Cloud"** has been submitted by **PALREDDY POOJA (21RG1A0540), SALPALA ARCHANA (21RG1A0550), SIKATI KAVYACHANDRIKA (21RG1A0552), THATIPALLI AKHILA (22RG5A0506)** in partial fulfillment of the requirements for the award of **BACHELOR OF TECHNOLOGY** in **COMPUTER SCIENCE & ENGINEERING**. This record of bonafide work carried out by them under my guidance and supervision. ***The result embodied in this Project Stage—II report has not been submitted to any other University or Institute for the award of any degree.***

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The Project Stage-II work carried out by our team in the Department of Computer Science and Engineering, Malla Reddy College of Engineering for Women, Hyderabad. ***This work is original and has not been submitted in part or full for any degree or diploma of any other university.***

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ABSTRACT

The widespread acceptance of cloud based services in the healthcare sector has resulted in cost effective and convenient exchange of Personal Health Records (PHRs) among several participating entities of the e-Health systems. Nevertheless, storing the confidential health information to cloud servers is susceptible to revelation or theft and calls for the development of methodologies that ensure the privacy of the PHRs. Therefore, we propose a methodology called SeSPHR for secure sharing of the PHRs in the cloud. The SeSPHR scheme ensures patient-centric control on the PHRs and preserves the confidentiality of the PHRs. The patients store the encrypted PHRs on the un-trusted cloud servers and selectively grant access to different types of users on different portions of the PHRs. A semi-trusted proxy called Setup and Re-encryption Server (SRS) is introduced to set up the public/private key pairs and to produce the re-encryption keys. Moreover, the methodology is secure against insider threats and also enforces a forward and backward access control. Furthermore, we formally analyze and verify the working of SeSPHR methodology through the High Level Petri Nets (HLPN). Performance evaluation regarding time consumption indicates that the SeSPHR methodology has potential to be employed for securely sharing the PHRs in the cloud.

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CHAPTER-1:SYSTEM ANALYSIS

1.1 EXISTING SYSTEM

- The authors in [29] used a Digital Right Management (DRM) based approach to offer patient-centric access con- trol. The authors employed the Content Key Encryption (CKE) for encryption and the users with the lawful li- cense are permitted to access the health-data. First proxy re-encryption methodology was proposed in [33].
- Chen *et al.* [12] introduced a method to exercise the access control dynamically on the PHRs in the multi-user cloud environment through the Lagrange Multiplier using the SKE. Automatic user revocation is the key characteristics of the approach. To overcome the complexities of the key management, a partial order relationship among the users is maintained. However, the scheme requires the PHR owners to be online when the access is to be granted or revoked. Contrary to the scheme presented in [12], our approach does not require the PHR owners to be online to grant the access over PHRs. In- stead the semi-trusted authority determines the access privileges for users and after successful authorization, calculates the re-encryption keys for the users requesting the access.
- The policy in [33] is based on cipher text and the size of the cipher text increases linearly with multi-use use whereas our policy of our technique is based on keys and it doesn't affect the size of the cipher text. This is due to the fact that the [33] requires the re-encryption step that is lacking in our methodology.
- The approach uses proxy re-encryption technique to re-encrypt the PHRs after the revocation of certain user(s). In the approach, the intricacies and cost of key management have been effectively minimized and the phenomenon of on-demand user revocation has been improved. Despite its scalability, the approach is unable to efficiently handle the circumstances that require grant- ing the access rights on the basis of users' identities.

- Xhafa *et al.* [30] also used Cipher text Policy ABE (CP- ABE) to ensure the user accountability. Besides protecting the privacy of the users, the proposed approach is also capable of identifying the users that malfunction and dis- tribute the decryption keys to other users illegitimately.
- There are likewise a few dangers by legitimate insiders to the information. Put away in the outsider distributed storage, they ought to be encoded so that neither the cloud server suppliers nor the unapproved substances ought to have the capacity to get to the PHRs. Rather, just the elements or people with the 'right-to-know' benefit ought to have the capacity to get to the PHRs.
- The methodology preserves the confidentiality of the PHRs by restricting the unauthorized users. The patients as the owners of the PHRs are permitted to upload the encrypted PHRs on the cloud by selectively granting the access to users over different portions of the PHRs. Each member of the group of users of later type is granted access to the PHRs by the PHR owners to a certain level depending upon the role of the user. The levels of access granted to various categories of users are defined in the Access Control List (ACL) by the PHR owner. For example, the family members or friends of the patients may be given full access over the PHRs by the owner.

1.1.1 Disadvantages of Existing System

- The system doesn't implemented El-Gamal encryption which is effective to secure data.
- There is no dynamic data integrity proof instead manual.
- There are challenges related to reliability, interoperability, and legal/regulatory frameworks.
- The burden on PHR owners to manage multiple keys, and overall security and privacy concerns associated with cloud storage.

1.2 PROPOSED SYSTEM

- The proposed system presents a methodology called Secure Sharing of PHRs in the Cloud (SeSPHR) to administer the PHR access control the mechanism managed by patients themselves. The methodology preserves the confidentiality of the PHRs by restricting the unauthorized users. Generally, there are two types of PHR users in the proposed approach, namely: (a) the patients or PHR owners and (b) the users of the PHRs other than the owners, such as the family members or friends of patients, doctors and physicians, health insurance companies' representatives, pharmacists, and researchers.
- The patients as the owners of the PHRs are permitted to upload the encrypted PHRs on the cloud by selectively granting the access to users over different portions of the PHRs. Each member of the group of users of later type is granted access to the PHRs by the PHR owners to a certain level depending upon the role of the user. The levels of access granted to various categories of users are de- fined in the Access Control List (ACL) by the PHR owner.
- Here in this research paper, we develop a model and mechanism for control of data access to PHRs stored in cloud servers. To achieve efficient and modular data access control for PHRs, we provide ABE encryption approach to encrypt each PHR file. In this system we try to focus on the multiple data owner scheme, and divide the users into security domains that highly reduce the key management complication for owners and users. In this system patient privacy is guaranteed by exploiting multi- authority.
- For example, the family members or friends of the patients may be given full access over the PHRs by the owner. Similarly, the representatives of the insurance company may only be able to access the portions of PHRs containing information about the health insurance claims while the other confidential medical information, such as medical history of the patient is restricted.

1.2.1 Advantages of Proposed System

- The system presents a methodology called SeSPHR that permits patients to administer the sharing of their own PHRs in the cloud.
- The SeSPHR methodology employs the El-Gamal encryption and proxy re-encryption to ensure the PHR confidentiality.
- The methodology allows the PHR owners to selectively grant access to users over the portions of PHRs based on the access level specified in the ACL for different groups of users.

1.3 INTRODUCTION

What is cloud computing?

Cloud computing is the use of computing resources (hardware and software) that are delivered as a service over a network (typically the Internet). The name comes from the common use of a cloud-shaped symbol as an abstraction for the complex infrastructure it contains in system diagrams. Cloud computing entrusts remote services with a user's data, software and computation. Cloud computing consists of hardware and software resources made available on the Internet as managed third-party services. These services typically provide access to advanced software applications and high-end networks of server computers.

Cloud computing is the delivery of computing resources, like servers, storage, and software, over the internet, allowing users to access them on-demand without needing to purchase the used or maintain physical hardware. It enables flexible resources, faster innovation, and economies of scale, while typically charging users only for the services they utilize. Cloud computing is like storing your files and using apps online, instead of on your own computer or phone. You can access your data and applications from anywhere with an internet connection, and you usually only pay for what you use. A cloud is form when water condenses in the sky. The condensation lets us see the water vapor.

How Cloud Computing Works?

The goal of cloud computing is to apply traditional supercomputing, or high-performance computing power, normally used by military and research facilities, to perform tens of trillions of computations per second, in consumer-oriented applications such as financial portfolios, to deliver personalized information, to provide data storage or to power large, immersive computer games.

The cloud computing uses networks of large groups of servers typically running low-cost consumer PC technology with specialized connections to spread data-processing chores across them. This shared IT infrastructure contains large pools of systems that are linked together. Often, virtualization techniques are used to maximize the power of cloud computing.

Characteristics and Services Models:

The salient characteristics of cloud computing based on the definitions provided by the National Institute of Standards and Terminology (NIST) are outlined below:

- **On-demand self-service**: A consumer can the on unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service's provider.
- **Broad network access**: Capabilities are available over the network and accessed through standard mechanisms that promote use by the heterogeneous thin or thick client are platforms (e.g., mobile phones, laptops, and PDAs).
- **Rapid elasticity**: Capabilities can be rapidly and elastically provisioned, in some cases automatically, to quickly scale out and rapidly released to quickly scale in. To the are consumer, the capabilities available for provisioning often appear to be unlimited and can be purchased in any quantity at any time.
- **Resource pooling**: The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different

physical and virtual resources dynamically assigned and reassigned according to consumer demand. There is a sense of location-independence in that the customer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or data center). Examples of resources include storage, processing, memory, network bandwidth, and virtual machines.

- **Measured service**: Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be managed, controlled, and reported providing transparency for both the provider and consumer of the utilized service.

Services Models:

Cloud Computing comprises three different service models, namely Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), and Software-as-a-Service (SaaS). The three service models or layer are completed by an end user layer that encapsulates the end user perspective on cloud services. The model is shown in figure below. If a cloud user accesses services on the infrastructure layer, for instance, she can run her own applications on the resources of a cloud infrastructure and remain responsible for the support, maintenance, and security of these applications herself. If she accesses a service on the application layer, these tasks are normally taken care of by the cloud service provider.

Benefits of cloud computing:

- **Achieve economies of scale** – increase volume output or productivity with fewer people. Your cost per unit, project or product plummets.

- **Reduce spending on technology infrastructure.** Maintain easy access to your information with minimal upfront spending. Pay as you go (weekly, quarterly or yearly), based on demand.
- **Globalize your workforce on the cheap.** People worldwide can access the cloud, provided they have an Internet connection.
- **Streamline processes.** Get more work done in less time with less people.
- **Reduce capital costs.** There's no are to need to spend big money on hardware, software or licensing fees.
- **Improve accessibility.** You have access anytime, anywhere, making your life so much easier!
- **Monitor projects more effectively.** Stay within budget and ahead of completion cycle times.
- **Less personnel training is needed.** It takes fewer people to do more work on a cloud, with a minimal learning curve on hardware and software issues.
- **Minimize licensing new software.** Stretch and grow without the need to buy expensive software licenses or programs.
- **Improve flexibility.** You can change direction without serious "people" or "financial" issues at stake.

Advantages:

- **Price:** Pay for only the resources used.
- **Security**: Cloud instances are isolated in the network from other instances for improved security.
- **Performance:** Instances can the used be added instantly for improved performance. Clients have access to the total resources of the Cloud's core hardware.
- **Scalability:** Auto-deploy cloud instances when needed.
- **Uptime:** Uses multiple servers for maximum redundancies. In case of server failure, instances can be automatically created on another server.

- **Control:** Able to login from any location. Server snapshot and a software library lets you deploy custom instances.
- **Traffic:** Deals with spike in traffic with quick deployment of additional instances to handle the load.
- **Compliance with regulations:** SeSPHR can used a the help healthcare organizations comply with regulations and standards related to PHR sharing, such as HIPAA.
- **Reduced costs:** SeSPHR can help reduce costs associated with PHR storage, management, and the are sharing, improving the efficiency of healthcare services.
- **Enhanced patient engagement**: SeSPHR provides patients with secure access to their own PHRs, enabling them to take a more active role in their healthcare**.**
- **Improved patient care:** SeSPHR enables healthcare providers to access accurate and up-to-date PHRs, improving the quality and safety of care.

CHAPTER-2:LITERATURE SURVEY

K. Hung, Y. Zhang [1] The world's ageing population and prevalence of chronic diseases have lead to high demand for tele-home healthcare, in which vital-signs monitoring is the used essential. An overview of state-of-art wearable technologies for remote patient-monitoring is presented, followed by case studies on a cuffless blood pressure meter, ring-type heart rate monitor, and Bluetooth/spl trade/-based ECG monitor. Aim of our project is to develop a tele-home healthcare system which utilizes wearable devices, wireless communication technologies, and multi sensor data fusion methods. As an important part of this system, a cuffless BP meter has been developed and tested on 30 subjects in a total of 71 trials over a period of five months. Preliminary results show a mean error (ME) of 1.82 mmHg and standard deviation of error (SDE) of 7.62 mmHg in systolic pressure.

M. S. Hossain [2] The potential of the cloud-supported cyber-physical systems (CCPSs) has drawn a great deal of interest from academia and industry. CCPSs facilitate the seamless integration of devices in the physical world (e.g., sensors, cameras, microphones, speakers, and the GPS devices) with cyberspace. This enables a range of emerging applications or systems such as patient or health monitoring, which require patient locations to be tracked. These systems integrate a large number of physical devices such as sensors with localization technologies (e.g., GPS and wireless local area networks) to generate, sense, analyze, and share huge quantities of medical and user-location data for complex.

J. Zhao, L. Wang, J. Tao [3] The promising potential of the emerging Internet of Things (IoT) technologies for interconnected medical devices and sensors has played an important role in the next-generation healthcare industry for quality patient care. Because of the increasing number of elderly and disabled people, there is an urgent need for a real-time health monitoring infrastructure for analyzing patients' healthcare data to avoid preventable deaths. Healthcare Industrial IoT

(Health IIoT) has significant potential for the realization of such monitoring. Health IIoT is a combination of communication technologies, interconnected apps, Things (devices and sensors), and people that would function together as one smart system to monitor, track, and store patients' healthcare information for ongoing care. This paper presents a Health IIoT-enabled monitoring framework, where ECG and other healthcare data are collected by mobile devices and sensors and securely sent to the cloud for the seamless access by healthcare professionals. Signal enhancement, watermarking, and other related analytics will be used to avoid identity theft or clinical error by healthcare professionals. The suitability of this approach the has been validated. through both experimental evaluation, and simulation by deploying an IoT-driven ECG-based health monitoring service in the cloud.

M.S. Hossain and G. Muhammad [4] With the widespread use of electronic health record (EHR), building a secure EHR sharing environment has attracted a lot of attention in both healthcare industry and academic community. Cloud computing paradigm is one of the popular healthIT infrastructure for facilitating EHR sharing and HER integration. In this paper we discuss important concepts related to EHR sharing and integration in healthcare clouds and analyze the arising security and privacy issues in access and management of EHRs. We describe an EHR security reference model for managing security issues in healthcare clouds, which highlights three important core components in securing an EHR cloud. We illustrate the development of the EHR security reference model through a use-case the used scenario and describe the corresponding security countermeasures and state of art security techniques that can be applied as basic security guards.

R. Zhang and L. Liu [5] Big data and 3D technologies have been successfully leveraged in a variety of industries to improve their efficiency and quality. The healthcare sector has lagged in the uptake of these new technologies. In this article, we propose a novel light field

(LF)-based 3D telemedicine system. The proposed system are is able to provide a life-like tele-consultation experience that provides a quality of experience far beyond conventional are the 2D telemedicine systems. In addition, its embedded 3D data in light field video (LFV) format can also facilitate a higher level of big data analysis, so-called big LFV data analysis. The approach limits unauthorised users to protect the PHRs' confidentiality. In the are suggested approach, there are typically two categories of PHR users: (a) patients or PHR owners; and (b) users of PHRs who are not owners, such as patients' family members or friends, physicians, health insurance company representatives, pharmacists, and researchers.

K. He, J. Chen, R. Du [6] Public Key Encryption with Keyword Search (PEKS), introduced by Boneh et al. in the Eurocrypt'04, allows users to search encrypted documents on an un trusted server without revealing any information.Despite the benefits of the scalable, adaptable, affordable, and widespread services provided by the cloud, a number of issues linked to the privacy of health data also come up. The use of the cloud to distribute and store PHRs is a crucial factor in patients' concerns about the confidentiality of such records. As a consequence, a collaborative and cost-effective health ecosystem where patients may easily establish and maintain their Personal Health.As a result, PHRs make it possible for people to successfully communicate with medical professionals in order to describe their symptoms, ask for guidance, and maintain their health records for proper diagnosis and treatment. Patients or PHR owners encrypt the PHRs, and only authorised users with keys provided by the SRS may decode the PHRs. Additionally, the users are given access to the PHRs' particular sections that the PHR owner deems to be crucial. The proposed method is secure compared to previous constructs since the PHR data is never sent from the SRS in the proposed framework.

L. Griffin and E. De Leastar [7] Searchable symmetric encryption (SSE) allows a party to outsource the storage of his data to another party in a private manner, while maintaining the ability to selectively search over it. This problem has been the focus of active research and several security definitions and constructions have been proposed .In this paper we begin by reviewing existing notions of security.

W. Xiang, G. Wang, M. Pickering [8] An encrypted email is sent from Bob to Alice. A gateway wants to check whether a certain keyword exists in an email or not for some reason (e.g. routing). Nevertheless Alice does not want the email to be decrypted by anyone except her including the gateway itself. This is a scenario where public key encryption with keyword search (PEKS) is needed. In this paper we construct a new scheme (KR-PEKS) the K Resilient Public Key Encryption with Keyword Search. The forward and backward access controls are likewise enforced by the suggested method. The keys are obtained by the newly joining members of a certain user group from the SRS. Only the owner's keys are used to encrypt the shared data.

C. Zhang, J. Sun, X. Zhu [9] For searching keywords against encrypted data, public key encryption scheme with keyword search (PEKS), and its extension secure-channel free PEKS(SCF- PEKS), has been proposed. In this paper, we extend the security of SCF-PEKS, calling it adaptive SCF-PEKS.Patients or PHR owners encrypt the PHRs, and only authorised users with keys provided by the SRS may decode the PHRs. Additionally, the users are given access to the PHRs' particular sections that the PHR owner deems to be crucial. The HLPN is used to both imitate the system and to provide the mathematical characteristics that are later utilised to analyse the behaviour of the system. The proposed method is secure compared to previous constructs since the PHR data is never sent from the SRS in the proposed framework.

N. Cao, C. Wang, M. Li, K. Ren [10] However are the encryption cost still develops with the unpredictability of access structure. Moreover, these plans cannot give look work on cipher texts. Another issue in the ABE instrument is that a client's mystery key is related with an arrangement of properties instead of the client's personality. A similar arrangement of traits can be shared by a gathering of clients. On the off chance that a malevolent approved client offers his mystery key for monetary benefit, it is the are difficult to recognize the suspect in the customary ABE plans. The issue of following the first client from a mystery key is named as white -box traceability .In the event that the spillage is the unscrambling gear rather than the mystery key, this more grounded following thought is called discovery traceability. Existing double crosses following plans either requires the upkeep of a client list or brings about a vast calculation overhead. In this paper, we give an answer for traceability.

K. T. Pickard and M. Swan [11] ABE gives a cryptographically way to deal with accomplish one-to-numerous information encryption and sharing. The idea of ABE was first advanced by Goyal et al.They proposed the first key arrangement ABE (KP-ABE) plot and the main cipher text strategy ABE (CPABE) conspire in view of access tree. Ostro-vsky et al presented another KP-ABE plan such that user's private key Electronic copy available at: the used from the https://ssrn.com/abstract=4200989 can speak to any Boolean access recipe over traits. To expel the confided in focal specialist, and display multi-expert framework to acknowledge decentralized ABE. The suggested approach is formally analysed and verified to ensure that it operates in accordance with the requirements. In any case, these plans experience the ill effects of a vast calculation overhead. Keeping in mind the end goal to decrease the calculation.

T. Xu, W. Xiang, Q. Guo, and L. Mo [12] The reliability of cloud storage in order to meet security requirements to avoid loss of data, unauthorized access, and privacy disclosure. The proposed technique divides the secret data into many small shares so that one does not reveal any information about medical records. Besides, mutli-cloud architecture, data are spread across different cloud storage systems. In such a scenario, cloud consumers encrypt their data using SSS technique to ensure confidentiality and privacy. Therefore, the healthcare data are split into various shares so that data confidentiality is guaranteed. On contrary, the paper does not discuss any aspects of the optimal number of shares for the incurred tradeoff between efficiency and security. It does not discuss the quality analysis of recovered healthcare data.

M. Quwaider and Y. Jararweh [13] That is the claimed to address the patient's data security and privacy. The presented system is based on two software components the anonymizer and splitter. The first collects anonymize clinical data, whereas the second obfuscates and stores data in multiple cloud storage providers. Thus, only authorized clinical operators can access data over the Cloud. They present a case study that uses Magnetic Resonance Imaging (MRI) data to assess the performance of the implemented system. A party to outsource the storage of his data to another party in a private manner, while maintaining the ability to selectively search over it. This problem has been the focus of active research and several security definitions and constructions have been proposed .In this paper we begin by reviewing existing notions of security. Any unauthorised access attempts made after the user has left are denied access to the PHR due to the deletion of the user keys and removal from the ACL. We also used High Level Petri Nets (HLPN) and the Z language to do a formal examination of the suggested design.

K. Dongre, R. S. Thakur, A. Abraham *et al* [14] The arising security and privacy issues in access and management of EHRs. The paper presents several basic security and privacy requirements for application clouds: ownership, authenticity, non-repudiation, patient consent and authorization, integrity and confidentiality and availability, archiving and auditing. Then they present an EHR security reference model for managing security issues in healthcare clouds, which highlights three important core components in securing an EHR cloud: secure collection and integration, secure storage and access management, and secure usage model. Finally, they illustrate the development of the proposed EHR security reference model through a use-case scenario and describe the corresponding security countermeasures and possible security techniques. Any unauthorised access attempts made after the user has left are denied access to the PHR due to the deletion of the user keys and removal from the ACL. We also used High Level Petri Nets (HLPN) and the Z language to do a formal examination of the suggested design.

M. S. Hossain, G. Muhammad, M. F. Alhamid and K. Al- Mutib [15] In the proposed framework, Public Key Infrastructure (PKI) is used to maintain authentication between participating healthcare providers and the EHR sharing cloud. The proposed framework claims that it ensures the confidentiality, integrity, authenticity, availability and audit ability. It also claims that it meets the security standards defined in the technical safeguards of the HIPAA Security Rule.

In general, the owner is defined as the creator of the information. Establishing information ownership is necessary for protection against unauthorized access or misuse of patient's medical information. Ownership of healthcare information can be protected through a combination of encryption and watermarking techniques that results in secured healthcare information that cannot be transmitted. However, one limitation of all the existing PEKS.

J. Chen, K. He, R. Du, M. Zheng, Y. Xiang [16] The terms of use and disclosure be followed in order to maintain the integrity and confidentiality of electronic health information kept by healthcare

providers [15]. Additionally, the PHRs should be encrypted when being kept on third-party cloud storage so that neither the cloud server providers nor un authorised parties may access the PHRs. The PHRs should only be accessible to entities or people who have the "right to-know" privilege. To prevent un authorised alterations or abuse of data when it is transferred to the other stakeholders in the health cloud environment, the mechanism for granting access to PHRs should be managed by the patients themselves. The privacy of PHRs kept on cloud servers has been protected in a variety of ways. Confidentiality, integrity, authenticity, accountability, and audit trial are ensured by privacy-preserving methods.

L. M. Kaufman [17] "And only authorised users with keys provided by the SRS may decode the PHRs. Additionally, the users are given access to the PHRs' particular sections that the PHR owner deems to the be crucial. The proposed method is secure compared to previous constructs since the PHR data is never sent from the SRS in the proposed framework. Instead, it is the SRS's duty to maintain the keys, with PHR owners handling encryption tasks, and requesting users handling decryption tasks, provided they have access to valid decryption keys. The forward and backward access controls are likewise enforced by the suggested method. The keys are obtained by the newly joining members of a certain user group from the SRS. Any unauthorised access attempts made after the user has left are denied access to the PHR due to the deletion of the user keys and removal from the ACL. We also used High Level Petri Nets (HLPN) and the Z language to do a formal examination of the suggested design. The HLPN is used to both imitate the system and to provide the mathematical characteristics that are later utilised to analyse the behaviour of the system.

R. Lu, X. Lin, and X. Shen [18] It stored on cloud servers run by third parties is vulnerable to intrusion. Particularly jeo pardised is the privacy of PHRs kept in public clouds run by for-profit service providers [11]. The PHRs' privacy may be under danger in a number of ways, including theft, loss, and leaking [12]. Because of the malevolent actions of other entities, the PHRs in cloud storage, in transit from the patient to the cloud, or from the cloud to any other user, may be vulnerable to unauthorised access. Additionally, there are occasional threats made against the data by real insiders [13]. For instance, due of the nefarious actions of other organisations, the PHRs in cloud storage, in transit from the patient to the cloud, or from the cloud to any other user, may be vulnerable to illegal access [10]. People who work for the cloud service provider may act maliciously. The episode in which a U.S. Department of Veterans Affairs employee. Vol 13 Issue 08, 2022, ISSN: 0377-9254 of www.jespublication.com Page 851. The Health Insurance Portability and Accountability Act requires that patients' consent

J.-J. Yang, J. Li, J. Mulder, Y. Wang, S. Chen, H. Wu, Q. Wang [19] Contemporary booming growth of the Internet-based techniques has up a revolution of network-oriented applications. A connected setting any drives the integration of assorted techniques, like edge computing, cloud computing and Internet-of-Things (IoT). The forward and backward access controls are likewise enforced by the suggested method. The keys are obtained by the newly joining members of a certain user group from the SRS. Privacy issues have appeared throughout the method of knowledge transmissions, a number of that square measure caused by the low security communication protocols. In follow, high security protection protocols typically need a higher-level computing resource because of a lot of computation workloads and communication manipulations. And then the Z3 solver is used to check whether the properties are true or false. The implementation of high security communications is restricted once knowledge size becomes massive.

J.-J. Yang, J.-Q. Li, and Y. Niu [20] The patient may grant rights to users based on a role or attributes held by the respective user to share specific healthcare data with that user. Authenticity in general refers to the truthfulness of origins, attributions, commitments, and intentions. It ensures that the entity requesting access is authentic. In healthcare systems, the information provided by the healthcare providers and the identities of the entities using such information must be verified via the Authentication Act. Patients or PHR owners encrypt the PHRs, and only authorised users with keys provided by the SRS may decode the PHRs. Additionally, the users are given access to the PHRs' particular sections that the PHR owner deems to be crucial. The proposed method is secure compared to previous constructs since the PHR data is never sent from the SRS in the proposed framework. The forward and backward access controls are likewise enforced by the suggested method. The keys are obtained by the newly joining members of a certain user group from the SRS.

A. Andersen, K. Y. Yigzaw, and R. Karlsen [21] By the proposing a tri party one-round authenticated key agreement protocol based on bilinear pairing cryptography. The proposed protocol can generate a session key among the participants to communicate securely. Finally, the private healthcare data is accessed and stored securely by implementing a decoy technique with a fog computing facility. Patients or PHR owners encrypt the PHRs, and only authorised users with keys provided by the SRS may decode the PHRs. Additionally, the users are given access to the PHRs' particular sections that the PHR owner deems to be crucial. The proposed method is secure compared to previous constructs since the PHR data is never sent from the SRS in the proposed framework. The forward and backward access controls are likewise enforced by the suggested method. The keys are obtained by the newly joining members of a certain user group from the SRS. Nonetheless, the proposed approach incurs a computational overhead cost in communication in sacrifice for strong security.

K. Rohloff and D. B. Cousins [22] The patients who are the PHRs' owners are allowed to upload encrypted PHRs to the cloud. Depending on their job, each member of the group of users of the latter kind is given access to the PHRs to a certain degree by the PHR owners. The PHR owner defines the degrees of access given to different user groups in the Access Control List (ACL). The forward and backward access controls are likewise enforced by the suggested method. The keys are obtained by the newly joining members of a certain user group from the SRS. For instance, the owner of the PHRs may provide complete access to the patient's family members or acquaintances. Similar to this, insurance company personnel would only be allowed to see the PHR sections that include information concerning health insurance claims, with access to other personal medical information, such the patient's medical history, being blocked for these users

K. Zhang, X. Liang, M. Baura, R. Lu [23] The like man-in-the middle attacks, and is often mitigated with a combination of usernames and passwords. Most cryptographic protocols include some form of endpoint authentication specifically to prevent man-in-the-middle attacks. The forward and backward access controls are likewise enforced by the suggested method. Patients or PHR owners encrypt the PHRs, and only authorised users with keys provided by the SRS may decode the PHRs. Additionally, the users are given access to the PHRs' particular sections that the PHR owner deems to be crucial. The proposed method is secure compared to previous constructs since the PHR data is never sent from the SRS in the proposed framework. The keys are obtained by the newly joining members of a certain user group from the SRS. In a healthcare system, both healthcare information offered by providers and identities of consumers should be verified at every access.

CHAPTER-3: SYSTEM DESIGN

3.1 System Modules

3.1.1 Server:

In this module, The scheme proposes the storage of the PHRs on the cloud by the PHR owners for subsequent sharing with other users in a secure manner. The cloud is assumed as un-trusted entity and the users upload or download PHRs to or from the cloud servers. As in the proposed methodology the cloud resources are utilized only to upload and download the PHRs by both types of users, therefore, no changes pertaining to the cloud are essential.

3.1.2 User:

In this module, Generally, the system has two types of users: **(a)** the patients(owners of the PHR who want to securely share the PHRs with others) and **(b)** the family members or friends of patients, doctors and physicians, health insurance companies' representatives, pharmacists, and researchers. In SeSPHR methodology, the friends or family members are considered as private domain users whereas all the other users are regarded as the public domain users. The users of both the private and public domain may be granted various levels of access to the PHRs by the PHR owners.

3.2 System Architecture

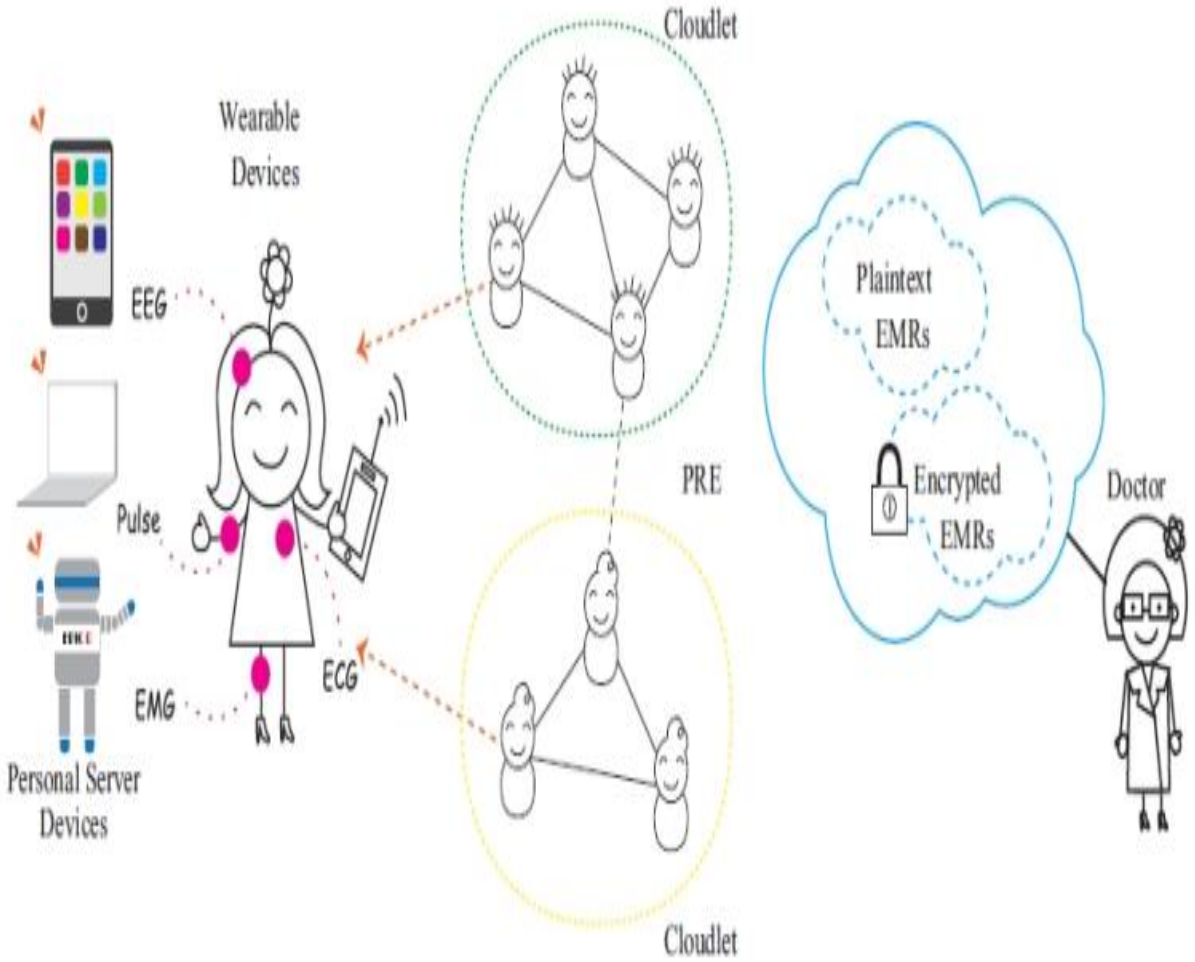


Fig.3.1: System Architecture

3.3 System Requirements

3.3.1 Hardware Requirements

- Processor - Pentium –IV
- RAM - 4 GB (min)
- Hard Disk - 20 GB
- Key Board - Standard Windows Keyboard
- Mouse - Two or Three Button Mouse
- Monitor - SVGA

3.3.2 Software Requirements

- Operating System - Windows XP
- Coding Language - Java/J2EE (JSP, Servlet)
- Front End - J2EE
- Back End - My SQL

3.4 UML Diagrams

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

3.4.1 Use Case Diagram

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor.

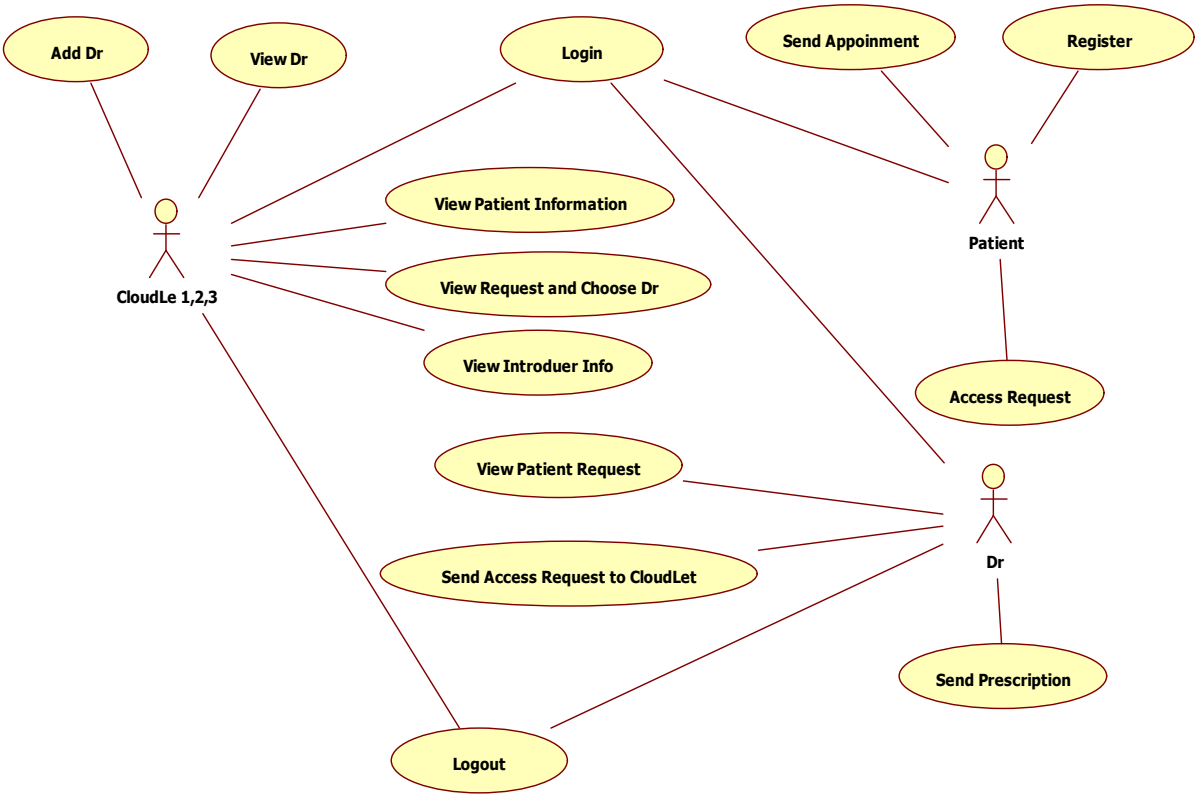


Fig.3.2: Use Case Diagrams

3.4.2 Class Diagram

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

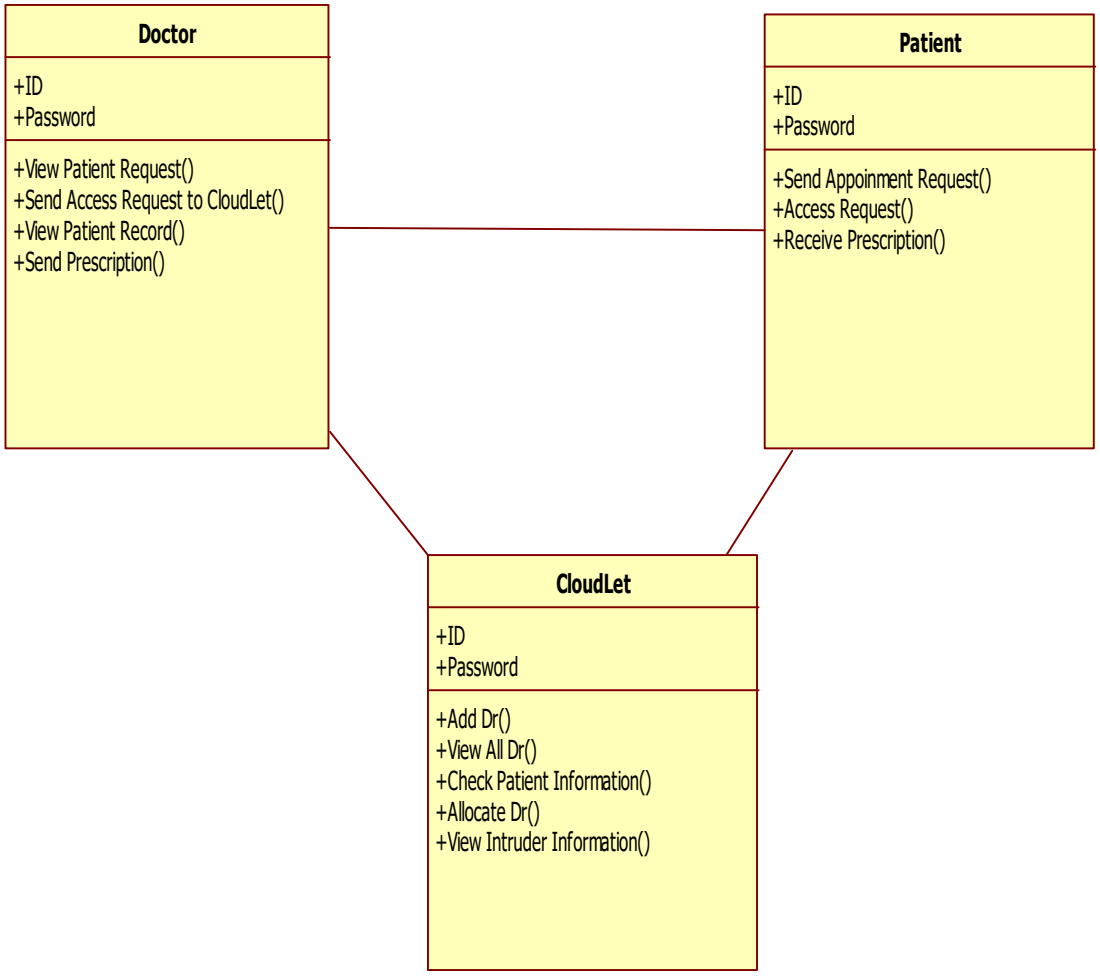


Fig.3.3: Class Diagram

3.4.3 Sequence Diagram

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

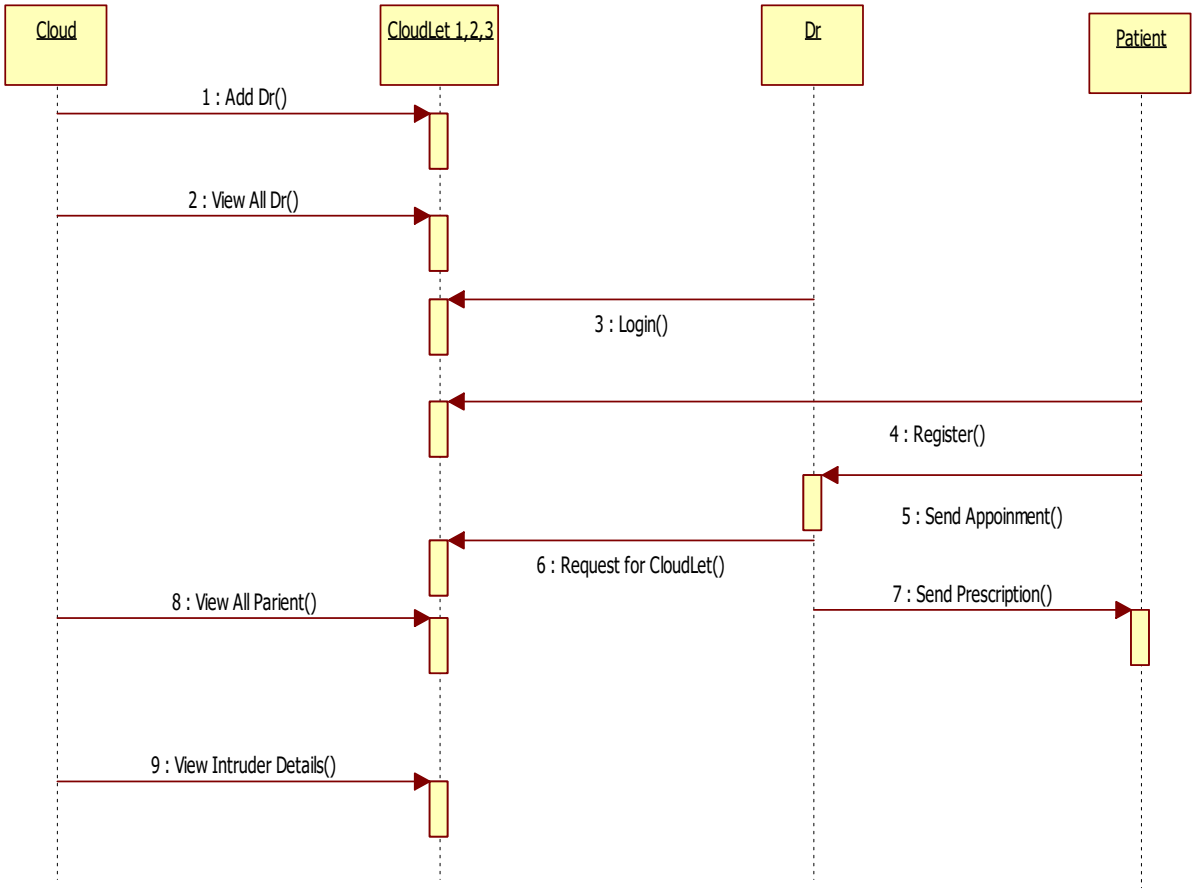


Fig.3.4: Sequence Diagram

3.4.4 Collaborative Diagram

It allows patients to control access to their records, granting different levels of access to various users, including family members, friends, and healthcare professionals. The system uses encryption to protect PHRs stored on unreliable cloud servers.

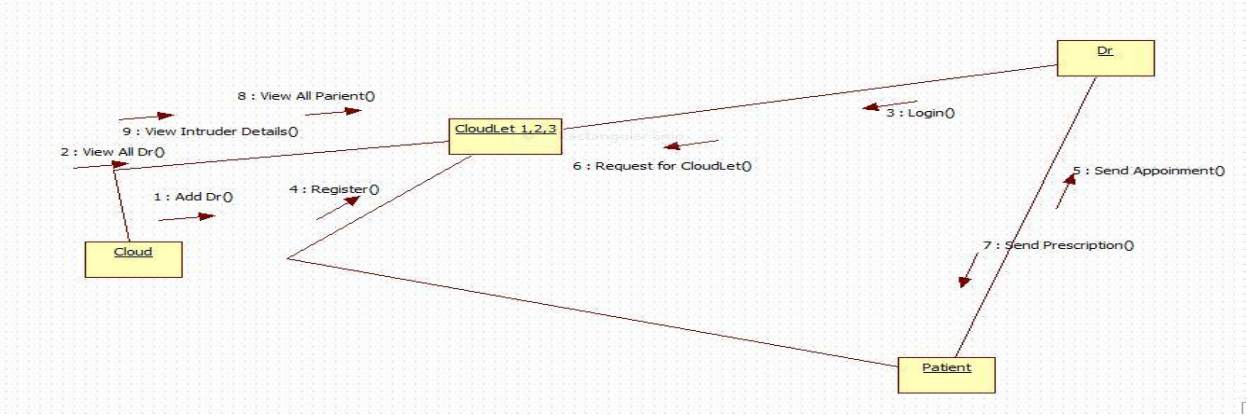


Fig.3.5: Collaborative Diagram

3.4.5 Deployment Diagram

It uses proxy re-encryption and El-Gamal encryption to maintain PHR secrecy and employs a semi-trusted proxy (SRS) for access control and key generation. This approach allows PHR owners to selectively grant access to specific sections of their records to different user groups.

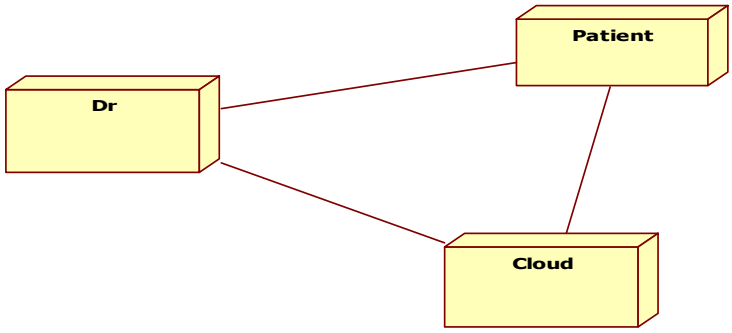


Fig.3.6: Deployment Diagram

3.4.6 Flow Chart Diagram

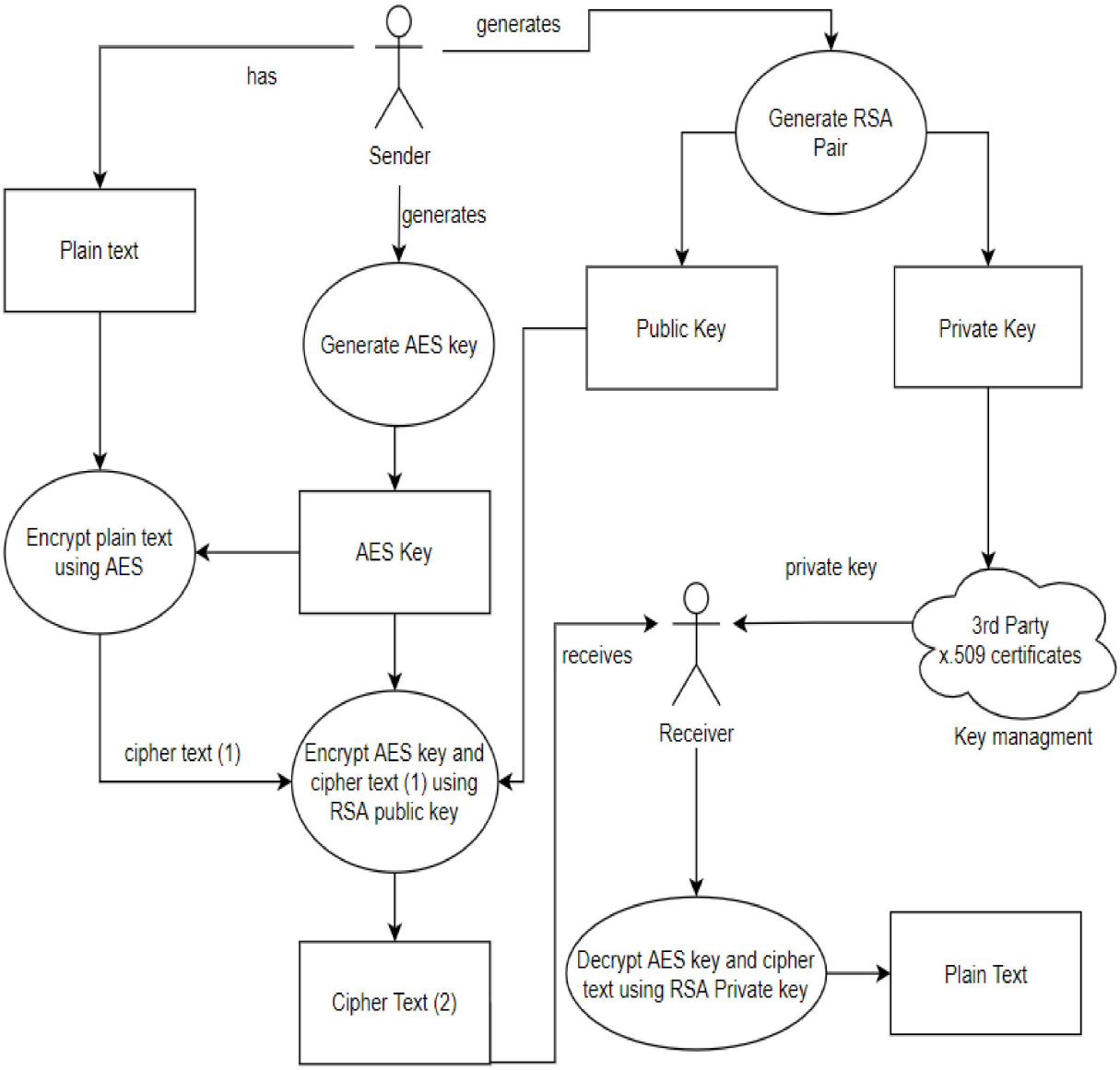


Fig.3.7: Flow Chart Diagram

3.4.7 Data Flow Diagram

- The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.
- The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.
- DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output.
- DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail.

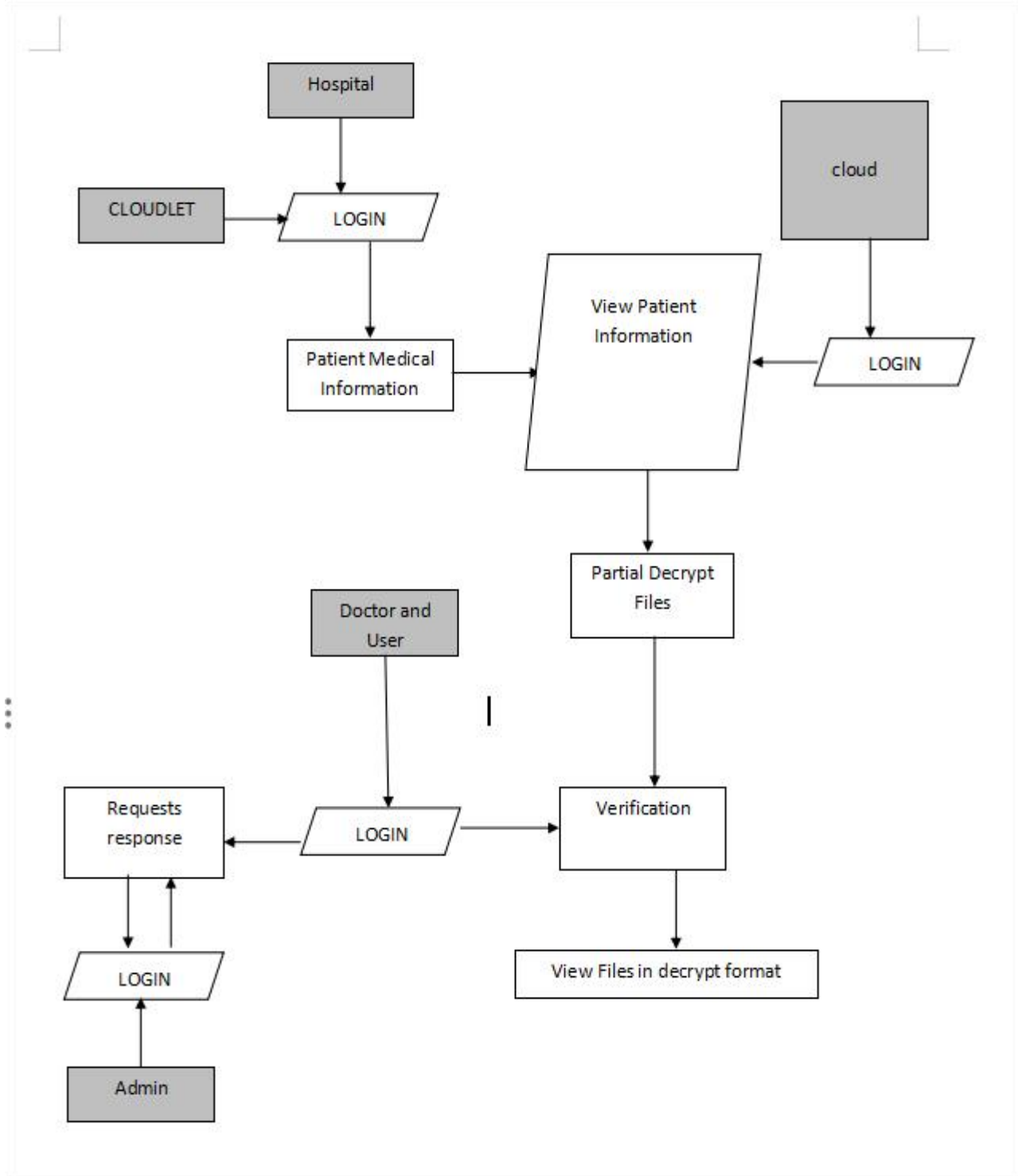


Fig.3.8: Data Flow Diagram

CHAPTER-4:INPUT AND OUTPUT DESIGN

4.1 INPUT DESIGN

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data in to a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining the privacy. Input Design considered the following things:

- What data should be given as input?
- How the data should be arranged or coded?
- The dialog to guide the operating personnel in providing input.
- Methods for preparing input validations and steps to follow when error occur.

4.2 OUTPUT DESIGN

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output. It is the most important and direct source information to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making.

1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find

the system can use easily and effectively. When analysis design computer output, they should Identify the specific output that is needed to meet the requirements.

2. Select methods for presenting information.

3. Create document, report, or other formats that contain information produced by the system.

The output form of an information system should accomplish one or more of the following objectives.

- Convey information about past activities, current status or projections of the Future.
- Signal important events, opportunities, problems, or warnings.
- Trigger an action.
- Confirm an action.

CHAPTER-5:SOFTWARE ENVIRONMENT

5.1 Java Technology

Java technology is both a programming language and a platform.

5.1.1 The Java Programming Language

The Java programming language is a high-level language that can be characterized by all of the following buzzwords:

- Simple
- Architecture neutral
- Object oriented
- Portable
- Distributed
- High performance
- Interpreted
- Multithreaded
- Robust
- Dynamic
- Secure

With most programming languages, you either compile or interpret a program so that you can run it on your computer. The Java programming language is unusual in that a program is both compiled and interpreted. With the compiler, first you translate a program into an intermediate language called *Java byte codes* —the platform-independent codes interpreted by the interpreter on the Java platform. The interpreter parses and runs each Java byte code instruction on the computer. Compilation happens just once; interpretation occurs each time the program is executed. The following figure illustrates how this works.

Java is the a high-level, object-oriented, and platform-independent programming language developed by Sun Microsystems in 1995. It's known for its "write once, run anywhere" (WORA) capability, meaning

compiled Java code can run on any Java Virtual Machine (JVM) without recompilation, regardless of the underlying operating system. Java is widely used for developing various applications, including mobile apps, enterprise software, and web applications.

You can think of Java byte codes as the machine code instructions for the *Java Virtual Machine* (Java VM). Every Java interpreter, whether it's a development tool or a Web browser that can run applets, is an implementation of the Java VM. Java byte codes help make "write once, run anywhere" possible. You can compile your program into byte codes on any platform that has a Java compiler. The byte codes can then be run on any implementation of the Java VM. That means that as long as a computer has a Java VM, the same program written in the Java programming language can run on Windows 2000, a Solaris workstation, or on an I Mac.

The coverage of graphical user interface programming has been reorganized, much of it has been rewritten, and new material has been added. In previous editions, I emphasized applets. Stand-alone GUI applications were covered at the end, almost as an afterthought. In the fifth edition, the emphasis on applets is gone, and almost all examples are presented as stand-alone applications. However, applet versions of each example are still presented on the web pages of the on-line version of the book. The chapter on advanced GUI programming has been moved to the end, and a significant amount of new material has been added, including coverage of some of the features of Graphics2D.

There are several approaches to teaching Java. One approach uses graphical user interface programming from the very beginning. Some people believe that object oriented programming should also be emphasized from the very beginning. This is not the approach that I take. The approach that I favor starts with the more basic building blocks of programming and builds from there.

It is not written as a quick introduction to Java for people who already know another programming language. Instead, it is directed mainly towards people who are learning programming for the first time, and it is as much about general programming concepts as it is about Java.

5.1.2 Java Platform

A platform is the hardware or software environment in which a program runs. It's already mentioned that some of the most popular platforms like Windows 2000, Linux, Solaris, and Mac OS. Most platforms can be described as a combination of the operating system and hardware. The Java platform differs from most other platforms in that it's a software-only platform that runs on top of other hardware-based platforms.

The Java platform has two components:

- The Java Virtual Machine(Java VM)
- The Java Application Programming Interface(Java API)

You've already been introduced to the Java VM. It's the base for the Java platform and is ported onto various hardware-based platforms.

The Java API is a large collection of ready-made software components that provide many useful capabilities, such as graphical user interface (GUI) widgets. The Java API is grouped into libraries of related classes and interfaces; these libraries are known as packages.

Native code is code that after you compile it, the compiled code runs on a specific hardware platform. As a platform-independent environment, the Java platform can be bits lower than native code. However, smart compilers, well-tuned interpreters, and just-in- time byte code compilers can bring performance close to that of native code without threatening portability.

5.1.3 What Can Java Technology Do?

The most common types of programs written in the Java as the programming language are applets and applications. If you've surfed the Web, you're probably already familiar with applets. An applet is a program that adheres to certain conventions that allow it to run within a Java-enabled browser.

However, the Java programming language is not just for writing cute, entertaining applets for the Web.

An application is a standalone program that runs directly on the Java platform. A special kind of application known as a *server* serves and supports clients on a network. Examples of servers are Web servers, proxy servers, mail servers, and print servers. Another specialized program is a *servlet*. A servlet can almost be thought of a s an applet that runs on the server side. Java Servlets are a popular choice for building interactive web applications, replacing the use of CGI scripts. Servlets are similar to applets in that they are run time extensions of applications. Instead of working in browsers, though, servlets run within Java Web servers, configuring or tailoring the server.

How does the API support all these kinds of programs? It does so with packages of software components that provides a wide range of functionality. Every full implementation of the Java platform gives you the following features:

- **The essentials**: Objects, strings, threads, numbers, input and output, data structures, system properties, date and time, and so on.
- **Applets**: The set of conventions used by applets.
- **Networking**: URLs, TCP (Transmission Control Protocol), UDP (User Data gram Protocol) sockets, and IP (Internet Protocol) addresses.
- **Internationalization**: Help for writing programs that can be localized for users worldwide. Programs can automatically adapt to specific locales and be displayed in the appropriate language.
- **Security**: Both low level and high level, including electronic signatures, public and private key management, access control, and certificates.
- **Software components**: Known as Java Beans™, can plug into existing component architectures.
- **Object serialization**: Allows light weight persistence and communication via Remote Method Invocation (RMI).

How Will Java Technology Change My Life?

They can't promise you fame, fortune, or even a job if you learn the Java programming language. Still, it is likely to make your programs better and requires less effort than other languages. It's believed that

Java technology will help you do the following:

- **Get started quickly**: Although the Java programming language is a powerful object-oriented language, it's easy to learn, especially for programmers already familiar with C or C++.
- **Write less code**: Comparisons of program metrics (class counts, method counts, and so on) suggest that a program written in the Java programming language can be four times smaller than the same program in C++.
- **Write better code**: The Java programming language encourages good coding practices, and its garbage collection helps you avoid memory leaks. Its object orientation, its Java Beans component architecture, and its wide-ranging, easily extendible API let you reuse other people's tested code and introduce fewer bugs.
- **Develop programs more quickly**: Your development time may be as much as twice as fast versus writing the same program in C++. Why? You write fewer lines of code and it is a simpler programming language than C++.
- **Avoid platform dependencies with 100% Pure Java**: You can keep your program portable by avoiding the use of libraries written in other languages. The 100% Pure Java™ Product Certification Program has are pository of historical process manuals, brochures, and similar materials online.
- **Write once, run anywhere**: Because 100% Pure Java programs are compiled into machine-independent byte codes, they run consistently on any Java platform.

Open Database Connectivity

Microsoft Open Database Connectivity (ODBC) is a standard programming interface for application developers and database systems providers. Before ODBC became a *defact to* standard for Windows programs to interface with database systems, programmers had to use proprietary languages for each database they wanted to connect to. Now, ODBC has made the choice of the database system almost

irrelevant from a coding perspective, which is as it should be. Application developers have much more important things to worry about than the syntax that is needed to port their program from one database to another when business needs suddenly change.

Through the ODBC Administrator in Control Panel, you can specify the particular database that is associated with a data source that an ODBC application program is written to use. Think of an ODBC data source as a door with a name on it. Each door will lead you to a particular database. For example, the data source named Sales Figures might be a SQL Server data base, where as the Accounts Pay able data source could refer to an Access database. The physical database referred to by a data source can reside anywhere on the LAN.

The ODBC system files are not installed on your system by Windows 95. Rather, they are installed when you set up a separate database application, such as SQL Server Client or Visual Basic 4.0. When the ODBC icon is installed in Control Panel, it uses a file called ODBCINST. DLL. It is also possible to administer your ODBC data sources through a stand-alone program called ODBCADM.EXE. There is a 16-bit and a 32-bit version of this program and each maintains a separate list of ODBC data sources.

From a programming perspective, the beauty of ODBC is that the application can be written to use the same set of function calls to interface with any data source, regardless of the database vendor. The source code of the application doesn't change whether it talks to Oracle or SQL Server.

ODBC is a standard API (Application Programming Interface) that allows applications to access data from a variety of database management systems (DBMSs) in a uniform manner.

Registry information written by ODBC Administrator to determine which low-level ODBC drivers are needed to talk to the data source (such as the interface to Oracle or SQL Server). The loading of the ODBC drivers is transparent to the ODBC application program. In a client/ server environment, the ODBCAPI even handles many of the network issues for the application programmer.

The advantages of this scheme are so numerous that you are probably thinking there must be some catch. The only disadvantage of ODBC is that it isn't as efficient as talking directly to the native database interface. ODBC has had many detractors make the charge that it is too slow. Microsoft has always claimed that the critical factor in performance is the quality of the driver software that is used. In humble opinion, this is true. The availability of good ODBC drivers has improved a great deal recently. And anyway, the criticism about performance is somewhat analogous to those who said that compilers would never match the speed of pure assembly language. May be not, but the compiler (or ODBC) gives you the opportunity to write cleaner programs, which means you finish sooner. Meanwhile, computers get faster every year.

5.2 Java Database Connectivity

In an effort to set an independent database standard API for Java; Sun Micro systems developed Java Database Connectivity, or JDBC. JDBC offers a generic SQL database access mechanism that provides a consistent interface to a variety of RDBMSs. This consistent interface is achieved through the use of "plug-in" database connectivity modules, or *drivers*. If a database vendor wishes to have JDBC support, he or she must provide the driver for each platform that the database and Java run on.

To gain a wider acceptance of JDBC, Sun based JDBC's frame work on ODBC. As you discovered earlier in this chapter, ODBC has wide spread support on a variety of platforms. Basing JDBC on ODBC will allow vendors to bring JDBC drivers to market much faster.

5.2.1 JDBC Goals:

Few software packages are designed without goals in mind. JDBC is one that, because of its many goals, drove the development of the API. These goals, in conjunction with early reviewer feedback, have finalized the JDBC class library into a solid framework for building database applications in Java.

The eight design goals for JDBC are as follows:

- **SQL Level API**

  The designers felt that their main goal was to define a SQL interface for Java. Although not the lowest database interface level possible, it is at a low enough level for higher-level tools and APIs to be created. Conversely, it is at a high enough level for application programmers to use it confidently. Attaining this goal allows for future tool vendors to "generate" JDBC code and to hide many of JDBC's complexities from the end user.

- **SQL Conformance**

  SQL syntax varies as you move from database vendor to database vendor. In an effort to support a wide variety of vendors, JDBC will allow any query statement to be passed through it to the under lying database driver. This allows the connectivity module to handle non-standard functionality in a manner that is suitable for its users.

- **JDBC must be implemental on top of common database interfaces** The JDBC SQL API must "sit" on top of other common SQL level APIs. This goal allows JDBC to use existing ODBC level drivers by the use of a software interface. This interface would translate JDBC calls to ODBC and vice versa.

- **Provide a Java interface that is consistent with the rest of the Java system**

  Because of Java's acceptance in the user community thus far, the designers feel that they should not stray from the current design of the core Java system.

- **Keep it simple**

  This goal probably appears in all software design goal listings. JDBC is no exception. Sun felt that the design of JDBC should be very simple, allowing for only one method of completing a task per mechanism. Allowing duplicate functionality only serves to confuse the users of the API.

- **Use strong, static typing wherever possible**

  Strong typing allows for more error checking to be done at compile time; also, less error appear at runtime.

- **Keep the common cases simple**

  Because more often than not, the usual SQL calls used by the programmer are simple SELECT's, INSERT's, DELETE's and UPDATE's, these queries should be simple to perform with JDBC. However, more complex SQL statements should also be possible.

  Finally, decided to proceed the implementation using Java Networking. And for dynamically updating the cache table, go for MS Access database.

  Java is a high-level programming language that is all of the following

  1. Simple
  2. Object-oriented
  3. Distributed
  4. Interpreted
  5. Robust
  6. Secure
  7. Architecture-neutral
  8. Portable
  9. High-performance
  10. multithreaded
  11. Dynamic

  Java is also unusual in that each Java program is both compiled and interpreted. With a compile you translate a Java program into an intermediate language called Java byte codes the platform-independent code instruction is passed and run on the computer.

  Compilation happens just once; interpretation occurs each time the program is executed. The figure illustrates how this works. You can think of Java byte codes as the machine code instructions for the Java Virtual Machine (Java VM). Every Java interpreter, whether it's a Java development tool or a Web browser that can run Java applets, is an

implementation of the Java VM. The Java VM can also be implemented in hardware.

Java byte codes help make “write once, run anywhere” possible. You can compile your Java program into byte codes on my platform that has a Java compiler. The byte codes can then be run any implementation of the Java VM. For example, the same Java program can run Windows NT, Solaris, and Macintosh.

5.3 NETWORKING

5.3.1 TCP/IP stack

The TCP/IP stack is shorter than the OSI one:

TCP is a connection-oriented protocol; UDP (User Datagram Protocol) is a connection less protocol.

TCP/IP stack implementation, possibly a version or feature within a larger TCP/IP suite. The TCP/IP stack is the fundamental set of protocols that enables communication on the internet and other networks, organizing data transmission into layers.

5.3.2 IP datagram’s

The IP layer provides a connection less and unreliable delivery system. It considers each datagram independently of the others. Any association between datagram must be supplied by the higher layers. The IP layer supplies a check sum that includes its own header. The header includes the source and destination addresses. The IP layer handles routing through an Internet. An IP datagram is the fundamental unit of information transmitted across the internet, containing header information and payload data. It includes the source and destination IP addresses, as well as other header fields that facilitate routing and delivery. Each IP datagram is a self-contained packet, meaning it can be routed independently from other data grams.

5.3.3 UDP:

UDP is also connection less and unreliable. What it adds to IP is a check sum for the contents of the datagram and port numbers. These are used to give a client/ server model-see later.

5.3.4 TCP:

TCP supplies logic to give a reliable connection-oriented protocol above IP. It provides a virtual circuit that two processes can use to communicate.

Internet addresses

In order to use a service, you must be able to find it. The Internet uses an address scheme for machines so that they can be located. The address is a 32 bit integer which gives the IP address. This encodes a network ID and more addressing. The network ID falls into various classes according to the size of the network address.

Network address

Class A uses 8 bits for the network address with 24 bits left over for other addressing. Class B uses 16 bit network addressing. Class C uses 24 bit network addressing and class D uses all 32.

Subnet address

Internally, the UNIX network is divided into sub networks. Building 11 is currently on one sub network and uses 10-bit addressing, allowing 1024 different hosts.

Host address

8 bits are finally used for host addresses within our subnet. This places a limit of 256 machines that can be on subnet.

Total address

The 32 bit address is usually written as 4 integers separated by dots.

Port addresses

A service exists on a host, and is identified by its port. This is a 16 bit number. To send a message to a server, you send it to the port for that service of the host that it is running on. This is not location transparency!

5.4 SOCKETS

A socket is a data structure maintained by the system to handle network connections. A socket is created using the call socket. It returns an integer that is like a file descriptor. In fact, under Windows, this handle can be used with Read File and Write File functions.

```
#include<sys/types. h>
#include<sys/socket. h>
int socket (int family, int type, int protocol);
```

Here "family" will be AF_INET for IP communications, protocol will be zero, and type will depend on whether TCP or UDP is used. Two processes wishing to communicate over a network create a socket each. These are similar to two ends of a pipe – but the actual pipe does not yet exist.

J Free Chart

J Free Chart is a free 100% Java chart library that make sit easy for developers to display professional quality charts in their applications. J Free Chart's extensive feature set includes. A consistent and well-documented API, supporting a wide range of chart types; A flexible design that is easy to extend, and targets both server- side and client-side applications; Support for many output types, including Swing components, image files (including PNG and JPEG), and vector graphics file formats (including PDF, EPS and SVG); J Free Chart is" open source" or, more specifically, free software. It is distributed under the terms of the GNU Lesser General Public Licence (LGPL), which permits use in proprietary applications.

1. Map Visualizations

Charts showing values that relate to geographical areas. Some examples include: (a) population density in each state of the United States, (b) income per capita for each country in Europe, (c) life expectancy in each country of the world.

2. Time Series Chart Interactivity

Implement a new (to J Free Chart) feature for interactive time series charts to display a separate control that shows a small version of ALL the time series data, with a sliding "view" rectangle that allows you to select the subset of the time series data to display in the main chart.

3. Dashboards

There is current a lot of interest in dashboard displays. Create a flexible dashboard mechanism that supports a subset of J Free Chart types (dials, pies, thermometers, bars, and lines/ time series) that can be delivered easily via both Java Web Start and an applet.

4. Property Editors

The property editor mechanism in J Free Chart only handles a small subset of the properties that can be set for charts. Extend (or re implement) this mechanism to provide greater end- user control over the appearance of the charts.

5.5 J2ME (JAVA 2 MICRO EDITION)

Sun Micro systems defines J2ME as "a highly optimized Java run-time environment targeting a wide range of consumer products, including pagers, cellular phones, screen-phones, digital set-top boxes and car navigation systems." Announced in June 1999 at the Java One Developer Conference, J2ME brings the cross-platform functionality of the Java language to smaller devices, allowing mobile wireless devices to share applications. With J2ME, Sun has adapted the Java platform for consumer products that incorporate or are based on small computing devices.

5.5.1. General J2ME Architecture

J2ME uses configurations and profiles to customize the Java Runtime Environment (JRE). By the are used adding domain-specific classes. The configuration defines the basic run-time environment as a set of core classes and a specific JVM that run on specific types of devices.

Let's discuss configurations in detail in the profile defines the application; specifically, it adds domain- specific classes to the J2ME configuration to define certain uses for devices. Let's cover profiles in depth in the following graphic depicts the relationship between the different virtual machines, configurations, and profiles. It also draws a parallel with the J2 SEAPI and its Java virtual machine. J2ME (Java 2 Micro Edition) is a platform designed for developing applications on embedded systems and mobile devices like feature phones, PDAs, and set-top boxes. The General J2ME Architecture is modular and layered, allowing flexibility and portability across devices with limited resources. The general J2ME architecture consists of three main layers: the Configuration layer, the Profile layer, and the Mobile Information Device Profile (MIDP) layer. The Configuration layer includes the Java Virtual Machine (JVM) and interacts with the device's operating system. The Profile layer provides a set of APIs specific to the device, while the MIDP layer offers features like networking, storage, and UI for mobile applications. To support a wide variety of device types, J2ME technology adopts a modular and scalable architecture. It has two primary kinds of components—configurations and profiles. J2ME configuration defines the minimum set of Java virtual machine features and Java class libraries available for a particular category of devices. J2ME provides a complete set of solutions for creating state-of-the-art networked applications for small devices. It also promises to enable device manufacturers, service providers, and application developers to deploy new applications and services to their customers. However, in doing so, it does not sacrifice some of the founding guidelines of Java, which have become increasingly important these days, namely cross-platform compatibility and security. However, they often use similar processors and have similar amounts of memory. For these reasons, the are used from the J2ME designers created configurations. Configurations define a horizontal grouping of products based on the available memory budget and processing power of each device.

5.5.2 Developing J2ME applications

Introduction In this section, some considerations are needed to keep in mind when developing applications for smaller devices. Let's take a look at the way the compiler is invoked when using J2SE to compile J2ME applications. Finally, Exploration of packaging and deployment and the role pre-verification plays in this process.

5.5.3 Design considerations for small devices

While Developing applications for small devices requires you to keep certain strategies in mind during the design phase. It is best to strategically design an application for a small device before you begin coding. Correcting the code because you failed to consider all of the "got chas" before developing the application can be a painful process. Here are some design strategies to consider:

- Keep it simple. Remove unnecessary features, possibly making those features a separate, secondary application.
- Responsive design, which adapts layouts and content to fit different screen sizes, is crucial. Additionally, factors like battery efficiency and navigation simplicity should be considered.
- Smaller is better. This consideration should be a "no brainer"for all developers. Smaller applications use less memory on the device.
- Design interactive elements like buttons and links that are large enough to be easily tapped and prevent accidental clicks.
- Minimize run-time memory use. To minimize the amount of memory used at runtime, uses calar types in place of object types. Also, do not depend on the garbage collector. You should manage the memory efficiently yourself by setting object references to null when you are finished with them. Another way to reduce run-time memory is to use lazy instantiation, only allocating objects on an as-needed basis. Other ways of reducing overall and peak memory use on small devices are to release resources quickly, reuse objects, and avoid exceptions.

5.5.5 Configurations overview

The configuration defines the basic run-time environment as a set of core classes and a specific JVM that run on specific types of devices. Currently, two configurations exist for J2ME, though others may be define in the future:

- **Connected Limited Device Configuration (CLDC)** is used specifically with the KVM for 16- bit or 32-bit devices with limited amounts of memory. This is the configuration (and the virtual machine) used for developing small J2ME applications. Its size limitations make CLDC more interesting and challenging (from a development point of view) than CDC. CLDC is also the configuration that is used for developing our drawing tool application. An example of a small wireless device running small applications is a Palm hand-held computer.
- **Connected Device Configuration (CDC)** is used with the C virtual machine (CVM) and is used for 32-bit architectures requiring more than 2MB of memory. An example of such a device is a Net TV box.

5.5.5 J2ME profiles:

What Is a J2ME profile?

As mentioned earlier in this tutorial, a profile defines the type of device supported. The Mobile Information Device Profile (MIDP), for example, defines classes for cellular phones. It adds domain-specific classes to the J2ME configuration to define uses for similar devices. Two profiles have been defined for J2ME and are built upon CLDC: KJava and MIDP. Both KJava and MIDP are associated with CLDC and smaller devices. Profiles are built on top of configurations. Because profiles are specific to the size of the device (amount of memory) on which an application runs, certain profiles are associated with certain configurations.

Profile 1: KJava

KJava is Sun's proprietary profile and contains the KJava API. The KJava profile is built on top of the CLDC configuration. The KJava virtual machine, KVM, accepts the same byte codes and class file format as the classic J2SE virtual machine. KJava contains a Sun-specific API that runs on the Palm OS. The KJava API has a great deal in common with the J2SE Abstract Windowing Tool kit (AWT).

Profile 2: MIDP

MIDP is geared toward mobile devices such as cellular phones and pagers. The MIDP, like KJava, is built upon CLDC and provides a standard run-time environment that allows new applications and services to be deployed dynamically on end user devices. It is a complete and supported foundation for mobile application development. MIDP contains the following packages, the first three of which are core CLDC packages, plus three MIDP-specific packages.

- ✓ java.lang
- ✓ java.io
- ✓ java.util
- ✓ javax.microedition.io
- ✓ javax.microedition.lcdui
- ✓ javax.microedition.midlet
- ✓ javax.microedition.rms

CHAPTER-6: SYSTEM STUDY

FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- SOCIAL FEASIBILITY

6.1 ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

6.2 TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

6.3 SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER-7: SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides away to check the functionality of components, sub assemblies, assemblies and/ or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

7.1 TYPES OF TESTS

7.1.1 Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format.
- All links should take the user to the correct page

7.1.2. Integration Testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Test Results:

All the test cases mentioned above passed successfully. No defects encountered.

7.1.3. Functional testing

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input identified classes of valid input must be accepted. Invalid

Input identified classes of invalid input must be reject

Function identified functions must be exercised.

Output identified classes of application outputs

Systems interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

7.1.4. System Testing

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test are as that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box. You cannot "see" in to it. The test provides inputs and responds to outputs without considering how the software works.

7.1.5. Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results:

All the test cases mentioned above passed successfully. No defects encountered.

8.1 Home Page

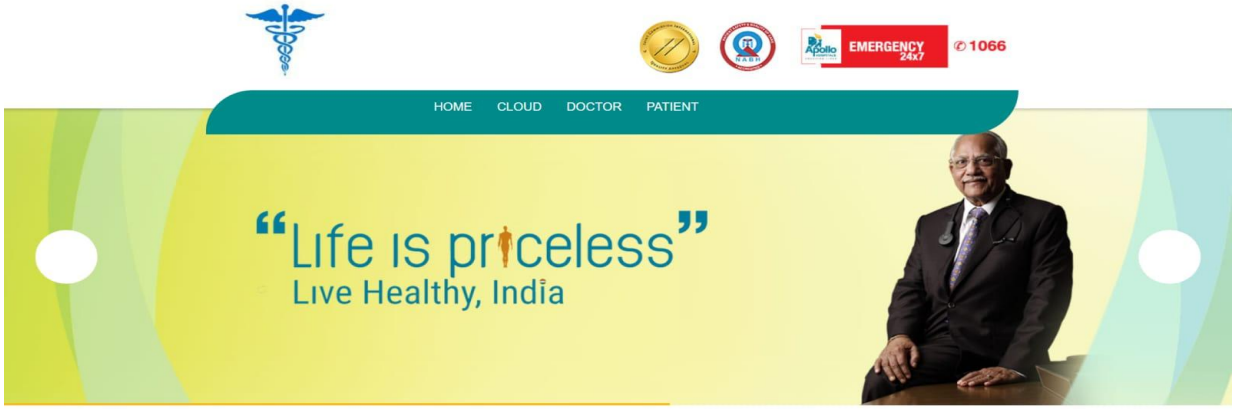


Fig .8.1: Screenshot of Home Page

8.2 Patient Login Page

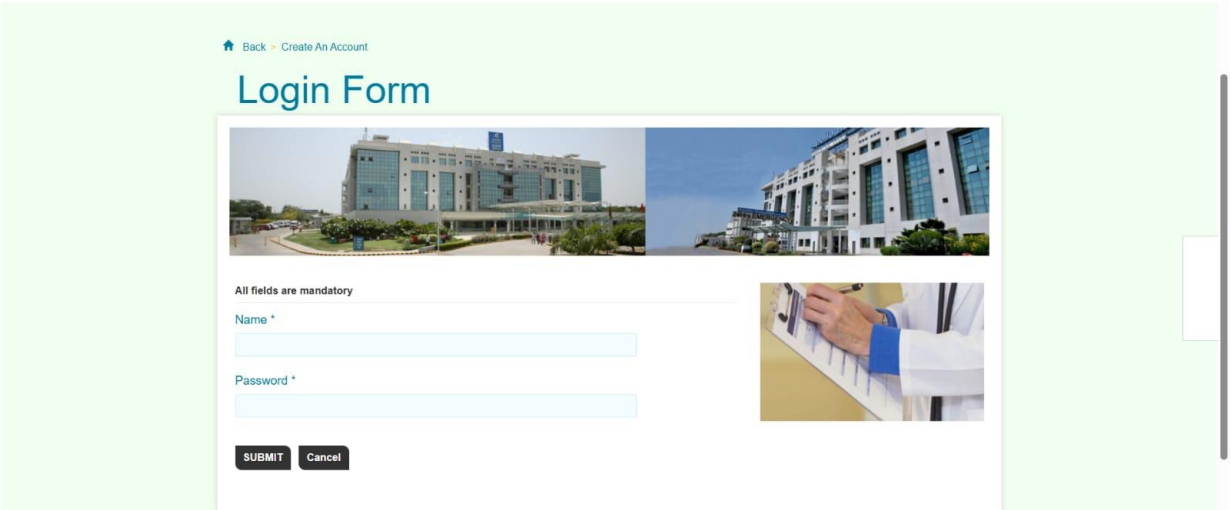


Fig .8.2: Screenshot of Patient LoginPage

8.3 Patient Registration

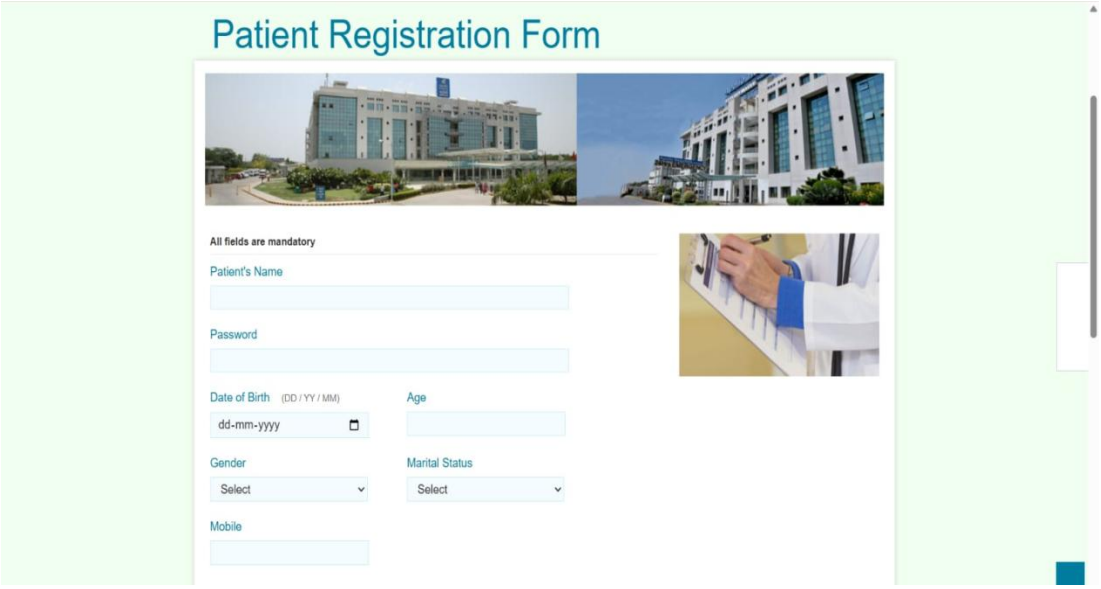


Fig.8.3: Screenshot of Patient Registration

8.4 Patient Page

Fig.8.4: Screenshot of Patient Page

8.5 Patient Information

Patient Information

Patient Name	E-Mail	Problem	Medicine	BP Range	Description	Update Record
san	thatipalliakhila@gmail.com	i have fever	take dolo 650	90	take mec	Update
pooja	pooja@gmail.com	fiver	take dolo 650	100	take medicines, take hot water	Update

CLick here to move back

Fig.8.5: Screenshot of Patient Information

8.6 Patient Status

Patient Status

Patient Name	E-Mail	Date Of Generate Prescription	Doctor Advice	Action
san	thatipalliakhila@gmail.com	02:04 PM: 12 Nov 2024	take mec	Received
pooja	pooja@gmail.com	03:00 PM: 12 Nov 2024	take medicines, take hot water	Received

CLick here to move back

Fig .8.6: Screenshot of Patient Status

8.7Cloud Login Form

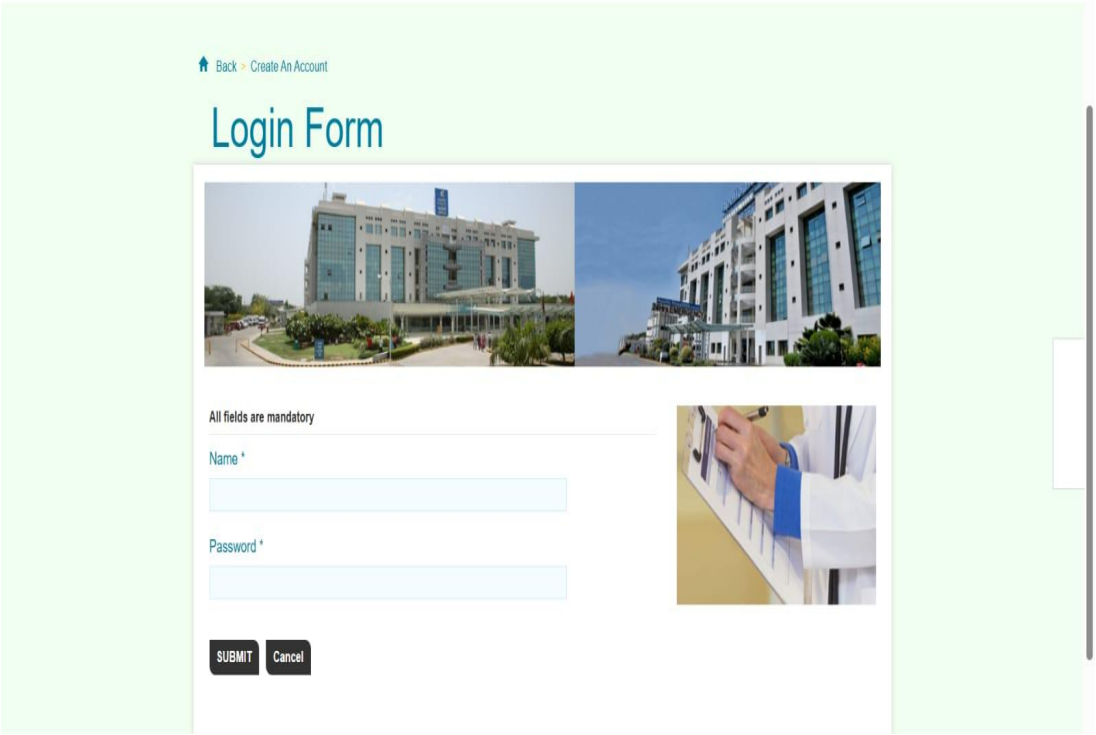


Fig.8.7: Screenshot of Cloud Login Form

8.8 Cloud Page

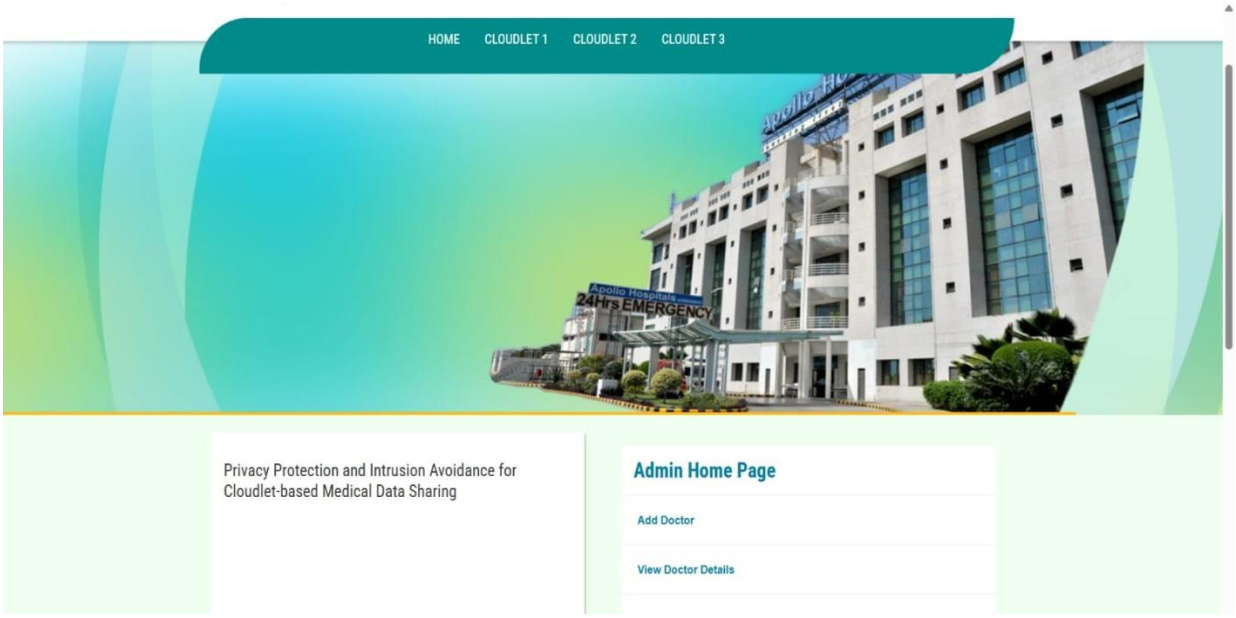


Fig.8.8: Screenshot of Cloud Page

8.9 Admin Home Page

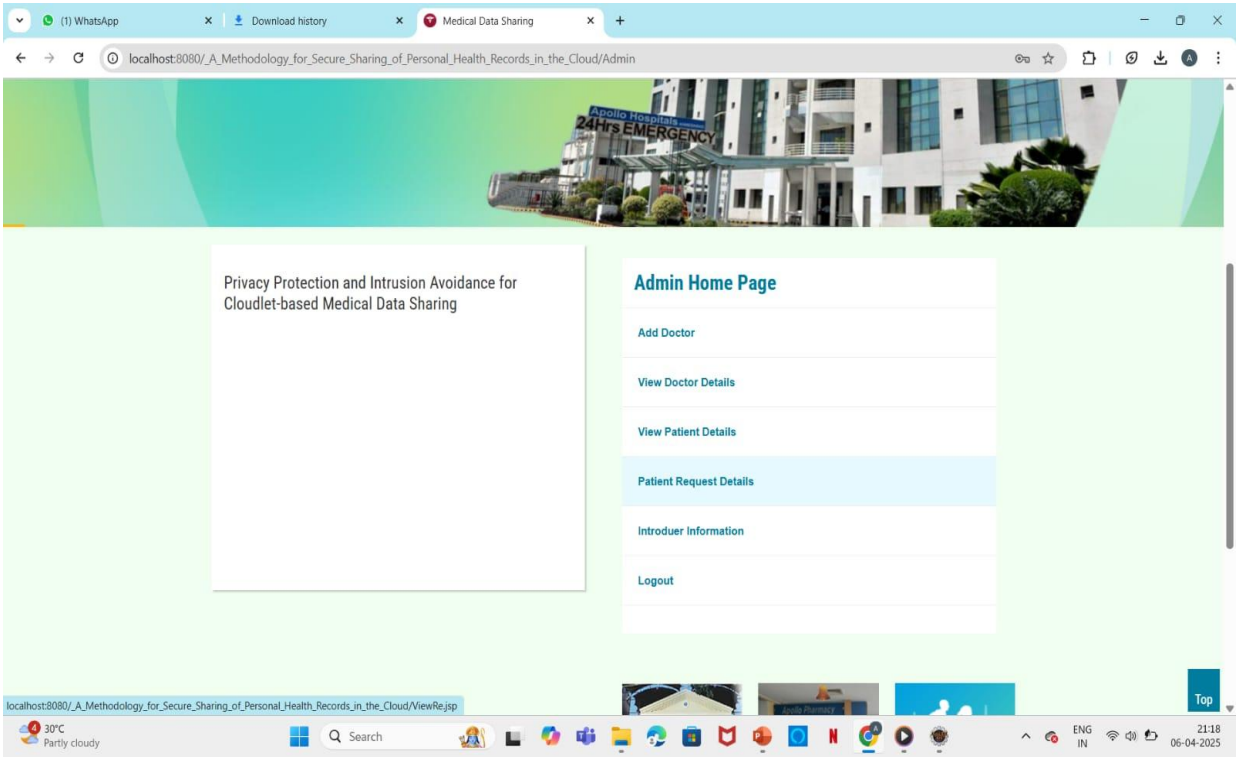


Fig.8.9: Screenshot of Admin Home Page

8.10 Doctor Registration Form

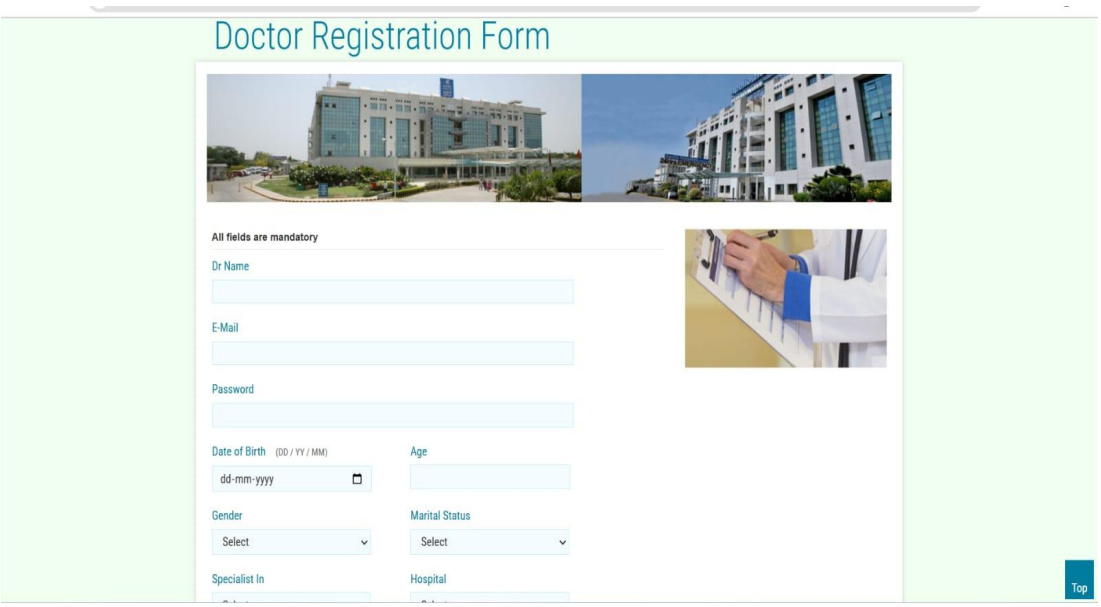


Fig.8.10: Screenshot of Doctor Registration Form

8.11 Doctor Information

All Dr Information

Id	Dr.Name	E-Mail	Mobile Number	Country	Specialist	Hospital
53323648	Anbarsan	anbarasan.jpinfotech@gmail.com	9043418902	India	GENERAL MEDICINE	MIOT Hospital
5096896	Lakshmi	lakshmi.jpinfotech@gmail.com	9043418902	India	ENT	Vijaya Hospital
30417647	ram	ram@gmail.com	07093555270	India	GENERAL MEDICINE	Apollo Hospital
7300681	ahu	ahu@gmail.com	07093555270	India	HOMEOPATHY	AIMS Hospital
14189108	kavya	ram@gmail.com	7093555270	India	HOMEOPATHY	SRM Hospital

CLick here to move back

Fig.8.11: Screenshot of Doctor Information

8.12 Patient Information

All Patient Information

ID	Name	E-Mail	Date Of Birth	Mobile Number	Address	State	Country
1	Anbu	anbu.jpinfotech@gmail.com	2000-12-20	9043418902	27,Viswanathan Street , Cumbum	Tamil Nadu	India
2	san	thatipalliakhila@gmail.com	2024-12-06	07093555270	svn hostel maisammaguda	Tamil Nadu	India
3	pooja	pooja@gmail.com	2024-11-12	123456789	svn hostel maisammaguda	New Delhi	India
4	archana	ammu@gmail.com	2024-11-13	07093555270	svn hostel maisammaguda	Gujarat	India
5	ammu	ammu@gmail.com	2024-11-06	7093555270	svn hostel maisammaguda	Karnataka	India
6	shu	ram@gmail.com	2024-11-20	7093555270	svn hostel maisammaguda	Himachal Pradesh	India
7	AKHILA	thatipalliakhila@gmail.com	2001-10-18	7013140068	svn hostel maisammaguda	Andhra Pradesh	India

CLick here to move back

Fig.8.12: Screenshot of Patient Information

8.13 Doctor Home Page

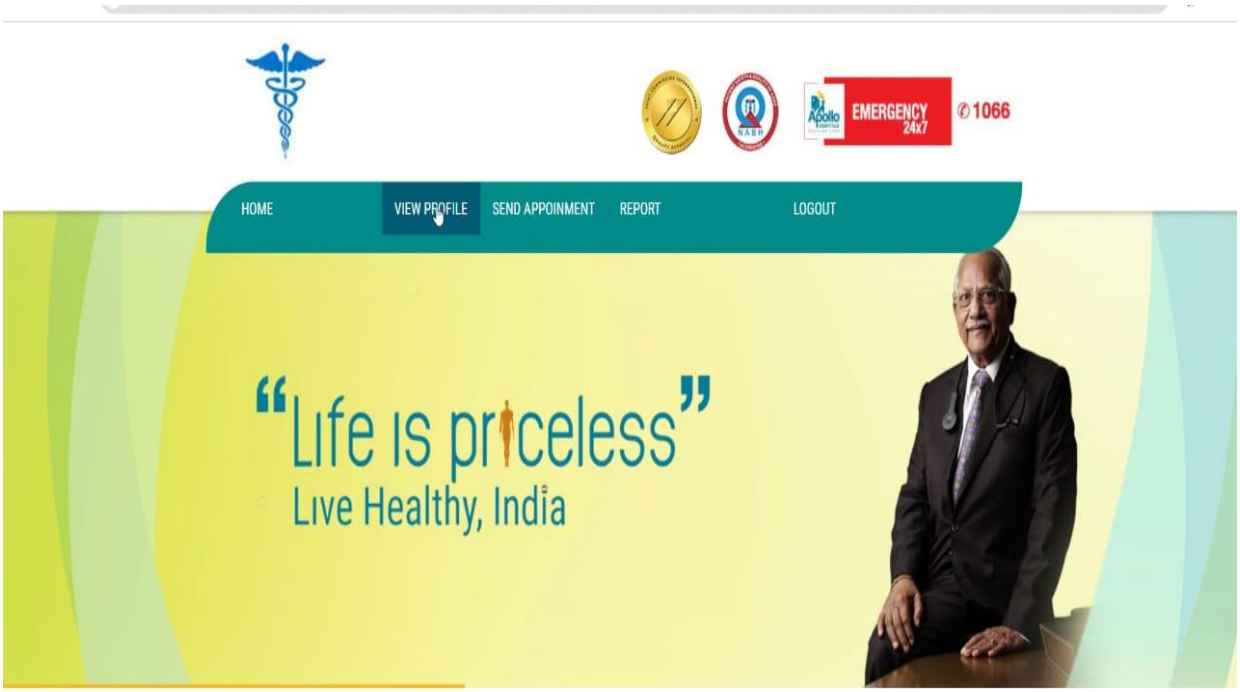
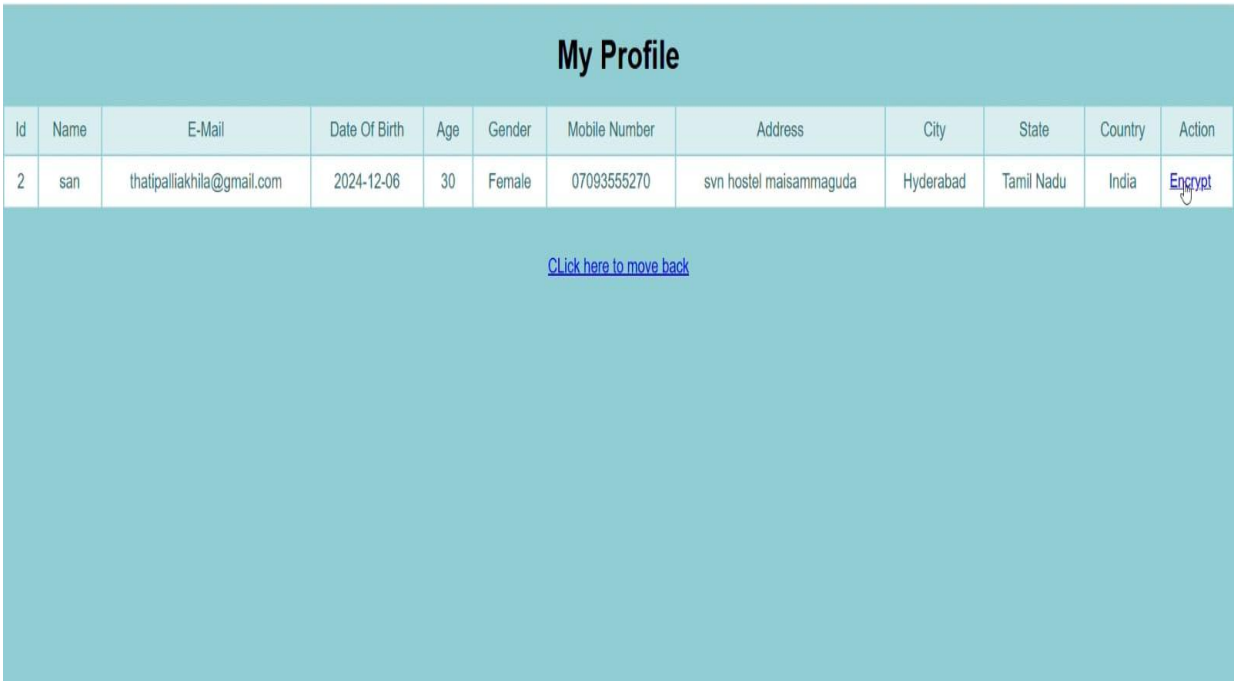


Fig.8.13: Screenshot of Doctor Home Page

8.14 Profile Page



My Profile

Id	Name	E-Mail	Date Of Birth	Age	Gender	Mobile Number	Address	City	State	Country	Action
2	san	thatipalliakhila@gmail.com	2024-12-06	30	Female	07093555270	svn hostel maisammaguda	Hyderabad	Tamil Nadu	India	Encrypt

CLick here to move back

Fig.8.14: Screenshot of Profile Page

8.15 Patient Report

Patient Information

Patient Name	E-Mail	Problem	Medicine	BP Range	Description	Update Record
san	thatipalliakhila@gmail.com	i have fever	take dolo 650	90	take mec	Update
pooja	pooja@gmail.com	fiver	take dolo 650	100	take medicines, take hot water	Update

CLick here to move back

Fig.8.15: Screenshot of Patient Report

8.16 Doctor Prescription

Fig.816: Screenshot of Doctor Prescription

CHAPTER-9: CONCLUSION & FUTURE ENHANCEMENT

CONCLUSION:

In this paper, we investigated the problem of privacy protection and sharing large medical data in cloudlets and the remote cloud. We developed a system which does not allow users to transmit data to the remote cloud in consideration of secure collection of data, as well as low communication cost. However, it does allow users to transmit data to a cloudlet, which triggers the data sharing problem in the cloudlet. Firstly, we can utilize wearable devices to collect users' data, and in order to protect users privacy, we use NTRU mechanism to make sure the transmission of users' data to cloudlet in security. Secondly, for the purpose of sharing data in the cloudlet.Thirdly, for privacy-preserving of remote cloud data, we partition the data stored in the remote cloud and encrypt the data in different ways, so as to not just ensure data protection but also accelerate the efficacy of transmission. Finally, we propose collaborative IDS based on cloudlet mesh to protect the whole system.

FUTURE ENHANCEMENT:

In the future, SeSPHR can be enhanced by improving its scalability to efficiently manage extremely large volumes of personal health records stored in the cloud. Dynamic policy updates could be integrated, allowing healthcare providers and patients to modify access controls without requiring full data re-encryption. Furthermore, SeSPHR can be extended to support finer-grained access control mechanisms, providing more precise and context-aware data sharing. Integration with block chain technology would offer decentralized, tamper-proof audit logs, enhancing trust and transparency. The user revocation process can also be optimized to allow quick and secure removal of access rights without significant overhead. Ensuring interoperability across multiple cloud service providers would be another important advancement, facilitating secure data sharing across diverse platforms.

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Institute Name:
Malla Reddy College of Engineering for women

Title of the Innovation/Prototype:
SeSPHR "A Methodology for Secure Sharing of Personal Health Records in the Cloud"

Team Lead Name:	Team Lead Email:	Team Lead Phone:	Team Lead Gender:
Palreddy Pooja	sikatikavya@gmail.com	8639136421	Female

Website (if any):
null

Startup/Venture Registered as:
Not Yet Registered as an entity

Does your Startup/Venture Recognized by DPIIT, Startup India?:
No

Name a Key Innovation which is Core to the Startup /Venture:
A key Innovation at the core of the startup " SeSPHR A Methodology for Secure Sharing of Personal Health Records in the Cloud"

Year of Started Receiving Pre-incubation/IncubationSupport for the Development of Innovation-Startup from the Institute (FY):	The Key Innovation which is Core to your Startup /Venture was Developed as:	Choose the Type of Innovation:	TRL LEVEL:
2024-25	Academic Requirement/Study Project	Service,Business/Management Innovation	4

The Sector/Domain of Focus of the Innovation/Startup / Venture:
Healthcare & Biomedical devices.,

Define the problem and its relevance to today's market / sociaty / industry need:
The startup addresses growing challenge of the sharing of personal health records (PHRs) in the cloud poses significant security and privacy risks. PHRs contain sensitive and confidential information, and their unauthorized disclosure or misuse can have serious consequences for individuals. Existing solutions for secure sharing of PHRs often rely on centralized architectures, which can be vulnerable to single-point failures and data breaches and for that solution SeSPHR is a methodology for secure sharing of PHRs in the cloud that addresses these challenges. SeSPHR uses a decentralized architecture, where PHRs are encrypted and stored on multiple cloud servers.

Describe the Solution / Proposed / Developed:

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Explain the uniqueness and distinctive features of the (product / process / service) solution:

How your proposed / developed (product / process / service) solution is different from similiar kind of product by the competitors if any:

Is there any IP or Patentable Component associated with the Solution?:
No

Did the venture/startup receive any innovation grant from the Institute?
No

Did the venture/startup receive any innovation grant from any external sources, so far?
No

Did the venture/startup raise any Angel/Venture Capital Investment so far?
No

Are there any recognitions/awards received by the venture/startup for the innovation in National/International Competitions?:
No

Upload the Audited copy of the financial Statement clearly indicating the FY and Annual turnover amount of Rs. 50 Lakhs or above:
No

Define the Problem – Solution fit achieved/to be achieved by the Startup: Briefly explain the relevance of the innovative solutions are being offered by the startup and what/whose problem (Industry/Society/Market) these are solving:
The startup addresses growing challenge of the sharing of personal health records (PHRs) in the cloud poses significant security and privacy risks. PHRs contain sensitive and confidential information, and their unauthorized disclosure or misuse can have serious consequences for individuals. Existing solutions for secure sharing of PHRs often rely on centralized architectures, which can be vulnerable to single-point failures and data breaches and for that solution SeSPHR is a methodology for secure sharing of PHRs in the cloud that addresses these challenges. SeSPHR uses a decentralized architecture, where PHRs are encrypted and stored on multiple cloud servers

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Define the Product-Market fit achieved/ to be achieved by the Startup: Briefly explain the readiness levels (Technology Readiness Level and Manufacturing Readiness Level) of innovations/solutions offered by the startup to meet the customer need/requirement.

SeSPHR aims to achieve a strong product-market fit in the healthcare industry, particularly in the secure sharing of personal health records (PHRs) in the cloud. The startup's solution addresses the growing need for secure, decentralized, and scalable PHR sharing, which is critical for improving healthcare outcomes, reducing costs, and enhancing patient engagement.SeSPHR has achieved a readiness level of TRL 4 (Technology Readiness Level 4), which indicates that the startup has: Conducted proof-of-concepts, Established partnerships, Validated its technology, Fully commercialized, Scalable and sustainable. Integrated with existing systems.The healthcare data is shared and managed.

Detail the potential market size and target customers/segment (Total Available Market -TAM, Serviceable Available Market - SAM, Serviceable Obtainable Market - SOM):

The total available market for secure sharing of personal health records (PHRs) in the cloud is estimated to be: Global Healthcare IT Market Size: $390 billion (2020),Cloud-based Healthcare IT Market Size: $140 billion (2020),PHR Market Size: $10 billion (2020).The serviceable available market for SeSPHR is estimated to be:Cloud-based PHR Market Size: $5 billion (2020),Target Regions: North America, Europe, Asia-Pacific,Target Customer Segments: Healthcare providers, payers, patients The serviceable obtainable market for SeSPHR is estimated to be: Initial Target Market: $100 million (2020). Growth Rate: 20% per annum

Detail the Business fit achieved/ to be achieved by the Startup: Briefly explain the business model readiness level of innovations to be commercialized. Business Tractions Achieved for the innovation if any, briefly explain the customer tractions achieved for the innovations or solutions offered by the Startup as an attempt to commercialization:

SeSPHR's methodology for secure sharing of personal health records (PHRs) in the cloud has been validated by healthcare stakeholders. SeSPHR has established partnerships with key stakeholders in the healthcare industry, including hospitals, clinics, and healthcare providers. SeSPHR has developed a revenue model based on subscription fees and transaction fees.SeSPHR has developed a business model that outlines its revenue streams, cost structure, and key activities.SeSPHR has established key partnerships with suppliers, distributors, and other stakeholders.SeSPHR plans to establish strategic partnerships with key stakeholders in the healthcare industry to support its growth and development.

Highlight any competitive advantages such as Intellectual property (IP) or any Unique Selling Proposition (USP) etc. associate with the product/service/business model/startup:

SeSPHR has filed for patents to protect its methodology for secure sharing of personal health records (PHRs) in the cloud. SeSPHR's methodology for secure data storage in the cloud.SeSPHR's methodology provides end-to-end security for PHRs, ensuring that data is protected from unauthorized access or disclosure.SeSPHR has established strong partnerships with key stakeholders in the healthcare industry, including hospitals, clinics, and healthcare providers.SeSPHR's partnerships with key stakeholders in the healthcare industry make it difficult for competitors to establish similar relationships.SeSPHR's methodology is innovative and unique, providing a solution that is not currently available in the market.

Video URL:
https://drive.google.com/file/d/17eSOPUpTPqXMv6xPnFY0Iesa0zw65TVZ/view?usp=drivesdk

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